CHAPTER

Underwater Acoustics

14

CHAPTER CONTENTS

14.1 Introduction 370
14.2 Sound propagation 370
14.2.1 Pressure 370
14.2.2 Intensity 371
14.2.3 Decibel 372
14.2.4 Transmission loss 372
14.2.4.1 Geometrical spreading 372
14.2.4.2 Absorption loss 373
14.2.4.3 One-way transmission loss 374
14.3 Transducers 374
14.3.1 Construction 374
14.3.2 Efficiency 374
14.3.3 Transducer bandwidth 375
14.3.4 Beam pattern 375
14.3.5 Directivity index 377
14.3.6 Transmitting response 378
14.3.7 Source level 379
14.3.8 Medium beam/narrow beam transducer 380
14.4 Acoustic noise 380
14.4.1 Environmental 380
14.4.2 Noise level calculations 380
14.4.3 Thruster noise 381
14.4.4 Sound paths 381
14.4.5 Sound velocity 381
14.4.6 Reflections 383

Mathematics is the language of science. All of the properties of underwater acoustics can be expressed in mathematical terms. Later in this chapter, as well as in Chapter 15 on sonar, two of the practical applications of underwater acoustics (acoustic positioning and imaging/profiling sonar) will be explored. In the initial sections of this chapter, the theoretical basis of underwater acoustics is presented to segue into the acoustic positioning and sonar applications.

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This section is fairly technical and is math based for those who would like to explore the theoretical aspects of underwater acoustics to gain a better insight into positioning and sonar. For those less interested in the technical background and more concerned with the practical applications of these technologies, proceed to Chapter 15 for sonar as well as Chapter 16 for acoustic positioning.

This chapter on underwater acoustics is based on a publication by Kongsberg Maritime entitled *Introduction to Underwater Acoustics*. A special thanks goes to Arndt Olsen with Kongsberg Maritime for obtaining permissions to include this material.

14.1 Introduction

Acoustic sound transmission represents the basic techniques for underwater navigation, telemetry, echo sounder, and sonar systems. Common for these systems are the use of underwater pressure wave signals that propagate with a speed of approximately 1500 m/s through the water (Figure 14.1).

When the pressure wave hits the sea bottom, or another object, a reflected signal is transmitted back and is detected by the system. The reflected signal contains information about the nature of the reflected object.

For a navigation and telemetry system, the communication is based upon an active exchange of acoustic signals between two or more intelligent units.

Transmission of underwater signals is influenced by a number of physical limitations, which together limit the range, accuracy, and reliability of a navigation or telemetry system.

The factors described in this section include:

- Transmitted power
- Transmission loss
- Transducer configuration
- Directivity and bandwidth of receiver
- Environmental noise
- Requirements of positive signal-to-noise ratio for reliable signal detection
- Ray bending and reflected signals.

The signal-to-noise ratio obtained can be calculated by the sonar equation.

14.2 Sound propagation

14.2.1 Pressure

A basic unit in underwater acoustics is pressure, measured in μPa (micropascal) or μbar. The Pa (Pascal) is now the international standard. It belongs to the MKS system, where 1 μPa = 10^6 N/m^2. The μbar belongs to the CGS system.

$$1\ \mu\text{bar} = 10^5\ \mu\text{Pa}$$

$$0\ \text{dB re } 1\ \mu\text{bar} = 100\ \text{dB re } 1\ \mu\text{Pa}$$

The μbar is a very small unit so negative decibels will rarely occur, if ever. To convert from μbar to μPa, simply add 100 dB.

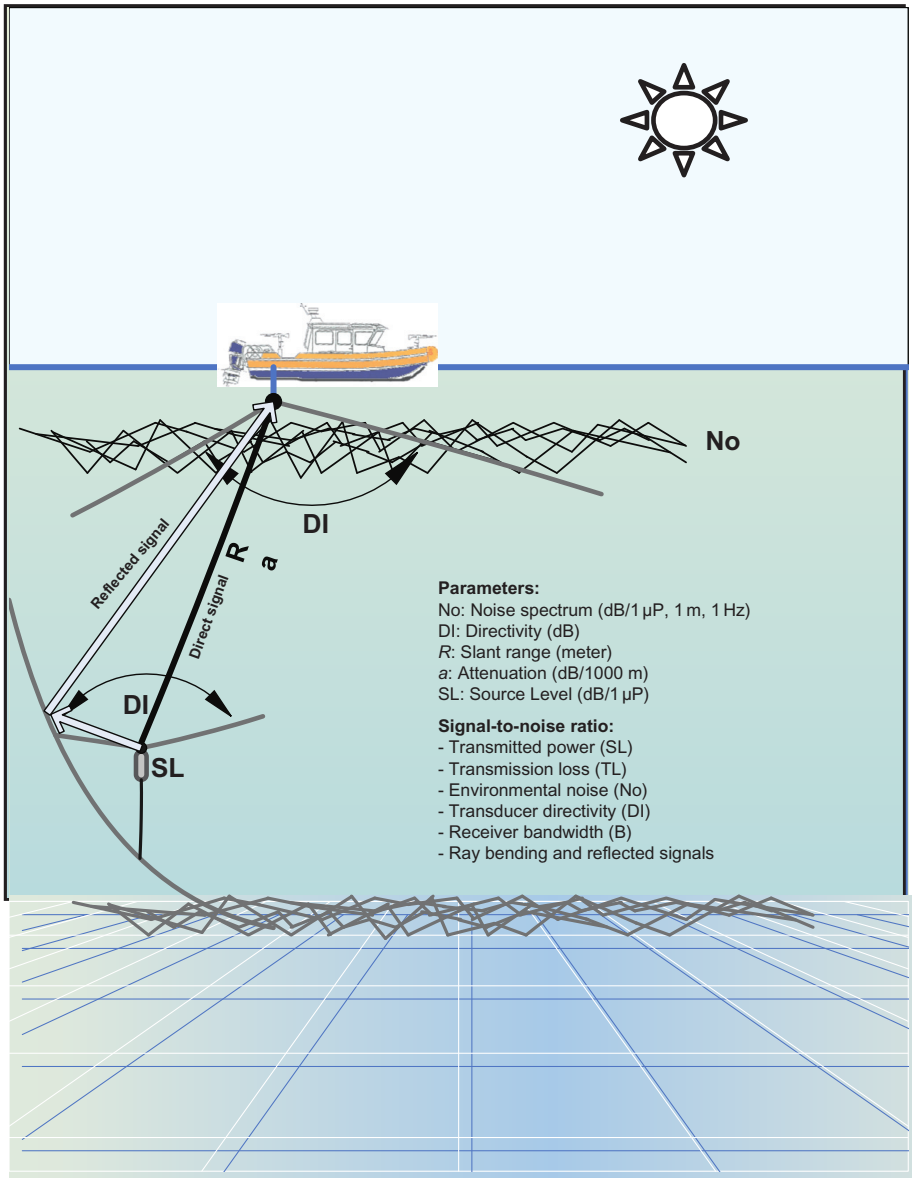


FIGURE 14.1

Parameters of the sonar equation.

14.2.2 Intensity

The sound intensity is defined as the energy passing through a unit area per second. The intensity is related to pressure by:

$$I = p^2/\rho c$$

where

I = intensity,
p = pressure,
ρ = water density, and
c = speed of sound in water.

14.2.3 Decibel

The decibel is widely used in acoustic calculations. It provides a convenient way of handling large numbers and large changes in variables. It also permits quantities to be multiplied together simply by adding their decibel equivalents. The decibel notation of intensity I is:

$$10 \log I/I_o$$

where I_o is a reference intensity.

The decibel notation of the corresponding pressure is:

$$10 \log(p^2/\rho c)/(\rho_o^2/\rho c) = 20 \log p^2/p_o$$

where p_o is the reference pressure corresponding to I_o.

Normally, p_o is taken to 1 μPa, and I_o will then be the intensity of a plane wave with pressure 1 μPa.

EXAMPLE

A pressure $p = 100$ μPa
In decibels: 20 log 100/1 = 40 dB re 1 μPa
The intensity will also be 40 dB re "the intensity of a plane wave with pressure 1 μPa."

As shown in the example, the decibel number is the same for pressure and intensity. It is therefore common practice to speak of sound level rather than pressure and intensity. The reference level is in both cases a plane wave with pressure 1 μPa.

14.2.4 Transmission loss

14.2.4.1 Geometrical spreading

When sound is radiated from a source and propagated in the water, it will be spread in different directions. The wave front covers a larger and larger area. For this reason the sound intensity decreases (with increasing distance from the source). When the distance from the source has become much larger than the source dimensions, the source can be regarded as a point source, and the wave front takes the form as a part of an expanding sphere. The area increases with the square of the distance (Figure 14.2) from the source, making the sound intensity decrease with the square of the distance.

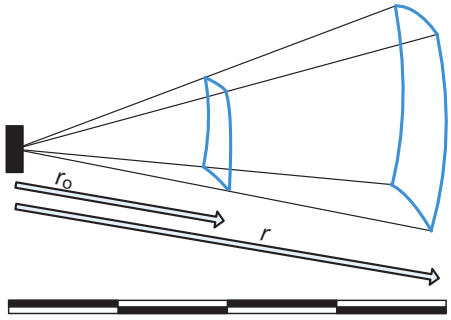


FIGURE 14.2

Transmission loss due to geometrical spreading.

Let I and I_o be the sound intensities in the distances r and r_o. Then:

$$I_o/I = (r/r_o)^2$$

Expressed in decibels, the geometrical spread loss is:

$$TL_1 = 10 \log I_o/I = 20 \log r/r_o$$

Usually a reference point is taken 1 m in front of the source. Setting $r_o = 1$ m we get:

$$TL_1 = 20 \log r$$

where r is measured in meters.

14.2.4.2 Absorption loss

When the sound propagates through the water, part of the energy is absorbed by the water and converted to heat. For each meter, a certain fraction of the energy is lost:

$$dI = -A \cdot I dr$$

where A is a loss factor. This formula is a differential equation with the solution:

$$I(r) = [I(r_o)/(e - Ar_o)] \cdot e^{-Ar}$$

$I(r_o)$ is the intensity at the distance r_o:

$$TL_2 = 10 \log I(r_o)/I(r) = \alpha(r - r_o)$$

where $\alpha = 10\, A \log (e)$.

Expressed in decibels, the absorption loss is proportional to the distance traveled. For each meter traveled, a certain number of decibels is lost.

If r_o is the reference distance 1 m, and if the range r is much larger than 1 m, the absorption loss will approximately be:

$$TL_2 = \alpha r$$

where α is named the absorption coefficient. Figure 14.3 shows absorption loss coefficient as a function of frequency. The value of α depends strongly on the frequency. It also depends on salinity, temperature, and pressure.

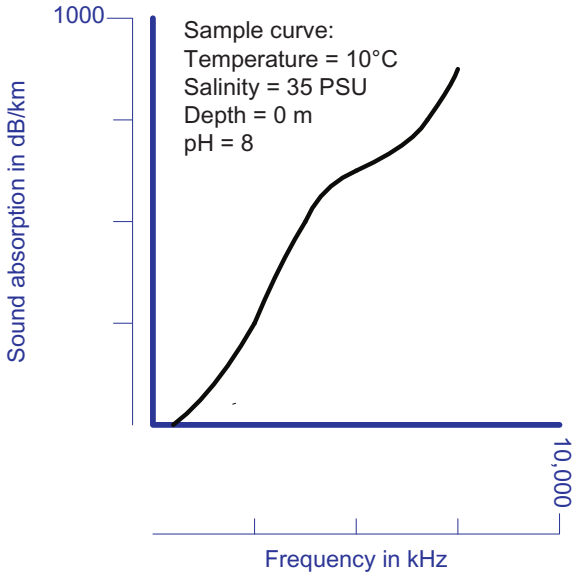


FIGURE 14.3

Absorption loss.

14.2.4.3 One-way transmission loss

The total transmission loss, which the sound suffers when it travels from the transducer to the target (Figure 14.4), is the sum of the spreading loss and the absorption loss:

$$TL = 20 \log r + \alpha r$$

where r is measured in meters and α is measured in dB/meter.

14.3 Transducers

14.3.1 Construction

A modern transducer is based on piezoelectric ceramic properties, which change physical shape when an electrical current is introduced (Figure 14.5). The change in shape, or vibration, causes a pressure wave, and when the transducer receives a pressure wave, the material transforms the wave into an electrical current. Thus, the transducer may act as both sound source and receiver.

14.3.2 Efficiency

When the transducer converts electrical energy to sound energy or vice versa, part of the energy is lost in friction and dielectric loss. Typical transducer efficiency is as follows:

- 50% for a ceramic transducer
- 25% for a nickel transducer

The efficiency is defined as the ratio of power out to power in.

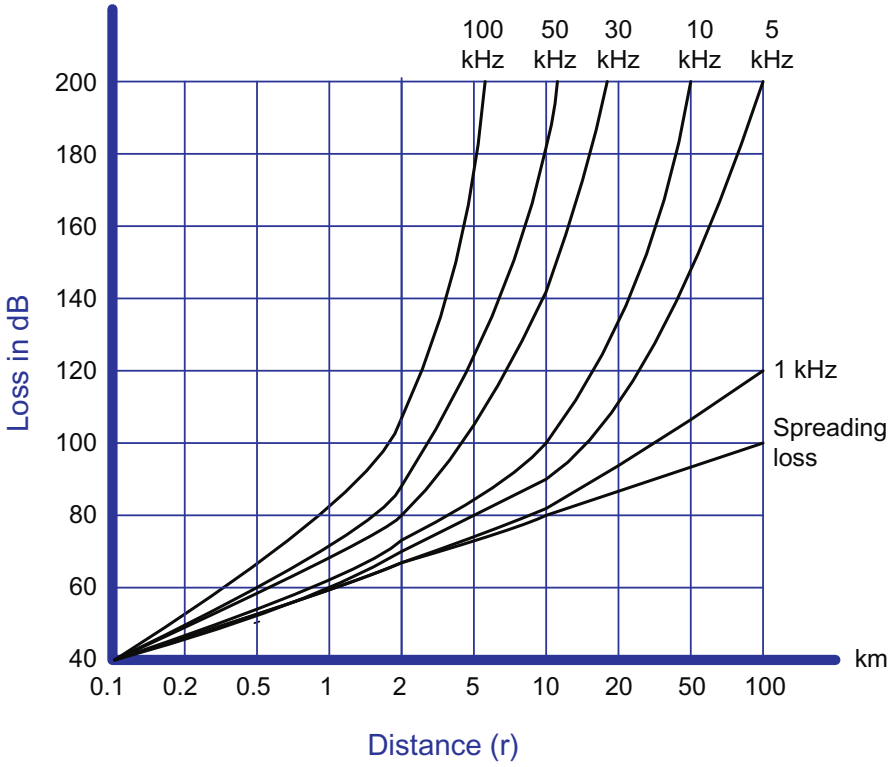


FIGURE 14.4

Propagation loss versus range (one-way transmission loss TL = 20 log $r = \alpha r$).

14.3.3 Transducer bandwidth

Normally a transducer is resonant. This means that they offer maximum sensitivity at the frequency they are designed for. Outside this frequency the sensitivity drops. Typically, the Q-value is between 5 and 10, where

$$Q = \text{center frequency}/\text{bandwidth (between 3 dB points)}$$

14.3.4 Beam pattern

The beam pattern shows the transducer sensitivity in different directions. It has a main lobe, normally perpendicular to the transducer face. The direction in which the sensitivity is maximum is called the *beam axis*. It also has unwanted side lobes and unwanted back radiation.

An important parameter is the beam width, defined as the angle between the two 3 dB points. As a rough rule of thumb, the beam width is connected with the size of the transducer by:

$$\beta = \lambda/L$$

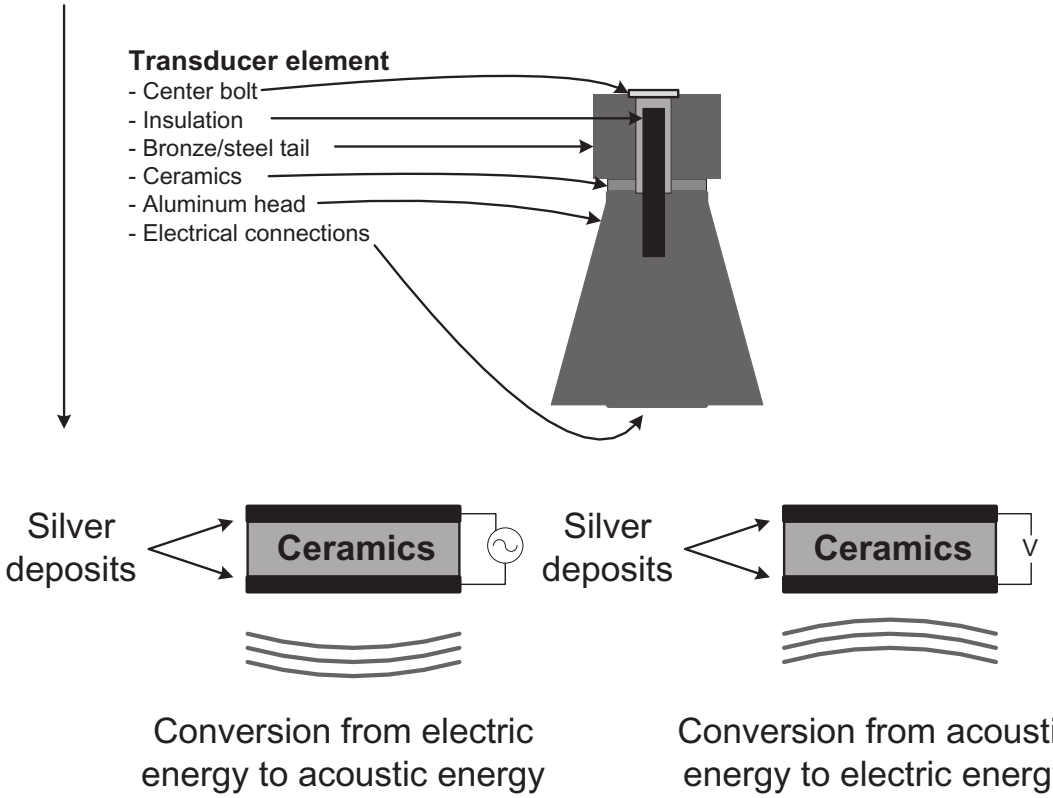


FIGURE 14.5

Cross-section of a typical ceramic transducer.

where

β = beam width in radians,
λ = wavelength,
L = linear dimension of the active transducer area (side for a rectangular area, diameter for a circular).

This rule is not valid for very small transducers, i.e., $L < \lambda$.

The theoretical beam pattern of a continuous line transducer is a sin *x*/*x* function, namely:

$$b(\theta) = (\sin(\pi L/\lambda \cdot \sin\theta)/\pi L/\lambda \cdot \sin\theta)^2$$

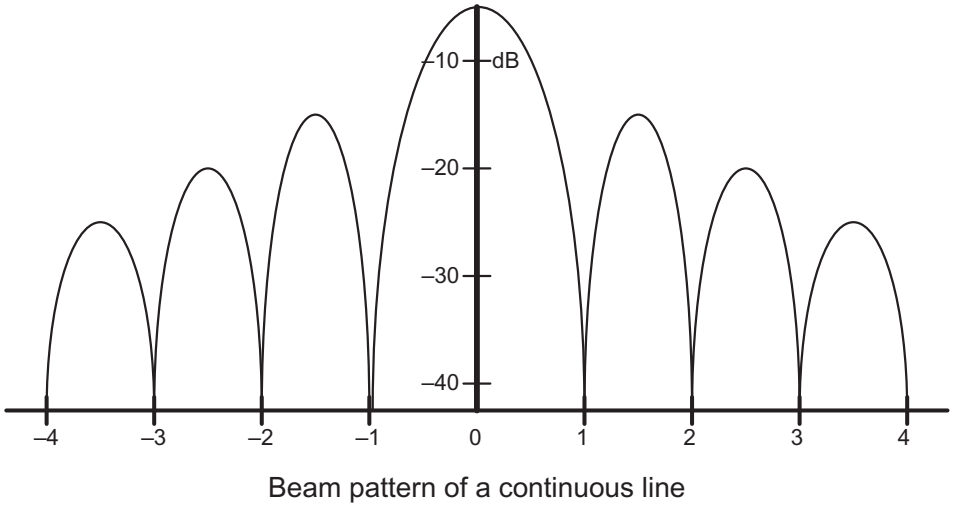


FIGURE 14.6

Beam pattern of a continuous line.

where

θ = angle from the beam axis,
L = line length,
b = transducer power response.

Figure 14.6 shows this pattern in a Cartesian plot. Note that the side lobes are gradually decreasing. The first side lobe is 13 dB below the point of maximum response.

A transducer having a rectangular active area vibrating uniformly as a piston will have this beam pattern in the two planes parallel to the sides.

In many transducers, the side lobes are reduced by a technique called "tapering."

14.3.5 Directivity index

With reference to transmission, the directivity index (DI) of a transducer can be defined by:

$$\mathrm{DI} = 10 \log(I_o/I_m)$$

where

I_o = the radiated intensity,
I_m = the mean intensity for all directions (including back radiation).

Both intensities are measured in the same distance from the transducer.

The mean intensity I is equal to the intensity we would get from an omnidirectional source, if this was given the same power and had the same efficiency as the transducer. We could therefore also define the DI as the ratio between the radiated intensity at the beam axis and the intensity an omnidirectional source would have given at the same point.

Table 14.1 DI Examples

Omnidirectional source	DI = 0 dB
Transducer with equal radiation everywhere in one half-plane and zero back radiation	DI = 3 dB
Typical echo sounder transducer	DI = 25 dB
Wide beam transducer	DI = 4 dB
Medium beam transducer	DI = 9 dB
Narrow beam transducer	DI = 15 dB
USBL Surface Station	DI = 25 dB

A transducer, which has a rectangular active area vibrating uniformly as a piston, will have this beam pattern in the two planes parallel to the sides. As shown in Table 14.1, the narrower the beam, the higher the DI.

The DI for a transducer with beam pattern $b(\theta, \Phi)$ and the mean intensity is found by integration over all directions, with solid angle element $d\Omega$, and division by the total solid angle 4π:

$$I_m = (1/4\pi)\int_{4\pi} I_o \cdot b(\theta, \Phi)d\Omega$$

According to the definition of DI:

$$DI = 10\log(4/\int_{4\pi} b(\theta, \Phi)d\Omega)$$

Calculation of DI after this formula is, however, no easy job, not even for the simplest transducer.

If the transducer side or diameter is larger than λ, the DI is approximately:

$$DI = 10\log(4\pi A/\lambda^2)$$

where A is the active transducer area, $A = L^2$.

When the beam width is known, another approximate formula can be used. For a rectangular transducer, the beam pattern in the two planes parallel to the sides are sin x/x function as mentioned previously.

The response is 3 dB down at:

$$(L/\lambda)\sin\theta_{3\ dB} = 0.443$$

Inserting this in the formula above gives:

$$DI = 10\log(2.47/[\sin(\beta_1/2)\cdot\sin(\beta_2/2)])$$

14.3.6 Transmitting response

The transmitting power response (S) of a transducer is the pressure produced at the beam axis 1 m from the transducer by a unit electrical input. The electrical input unit may be volt, ampere, or watt. A typical value for the transmitting response for a ceramic transducer is:

$$S = 193\ \text{dB re } 1\ \mu\text{Pa}/W$$

14.3.7 Source level

The source level (SL) of sonar or an echo sounder is the sound pressure in the transmitted pulse at the beam axis 1 m from the transducer. If the transmitting response (S) is known, then the source level is:

$$\mathrm{SL} = S + 10 \log P$$

where

P = transmitter power,
S = transmitting power response.

A widely used formula is:

$$\mathrm{SL} = 170.9 + 10 \log P + E + \mathrm{DI}$$

where

SL = source level in dB re 1 μPa,
P = transmitter power in watts,
$E = 10 \log \eta$,
η = transducer efficiency,
DI = directivity index.

The constant 170.9 incorporates conversion from watts to pascals and can be derived as follows:

$$\mathrm{SL} = 10 \log(1/4\pi) + 10 \log P + E + \mathrm{DI}$$

The factor $1/4\pi$ represents the source level from an omnidirectional source, supplied with 1 W electrical power and with 100% efficiency. When 1 W sound power is distributed over a sphere with radius 1 m and surface 4π m^2 the sound intensity will be:

$$(1/4\pi)\ W/m^2$$

The connection between pressure and intensity is:

$$p = \sqrt{I \rho c}$$

where

I = intensity,
ρ = density of water,
p = pressure,
c = sound velocity (in water).

Seawater at temperature 10°C with salinity 35 (PPT) at the sea surface has the following values:

$\rho = 1027$ kg/m^3
$c = 1490$ m/s

A sound density of 1/4 W/m^2 will in this environment correspond to a sound pressure of:

$$p = 349 \times 10^6\ \mu\mathrm{Pa} = 170.9\ \mathrm{dB\ re\ 1\ \mu Pa}$$

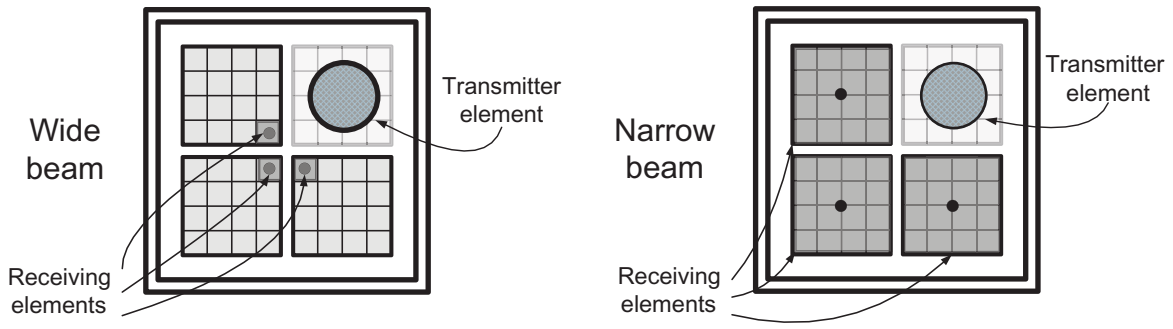


FIGURE 14.7

Multielement transducers.

14.3.8 Medium beam/narrow beam transducer

The transducer used in USBL (ultrashort baseline—Europeans use the term SSBL or "super short baseline") mode (acoustic positioning) normally consists of three different groups of elements. This is to be able to calculate a three-dimensional bearing to the transponders.

- The beam width of a transducer can be changed during operation. This is achieved by combining more transducer elements in series/parallel.
- A more narrow beam gives higher directivity (higher gain and higher noise suppression from outside the beam) but will give a smaller signal "footprint."
- For the *narrow beam transducer* in Figure 14.7, the 3 dB point in wide beam mode is 160°, while in narrow beam mode at the 3 dB point it is 30°.
- When using the transducers in SBL (short baseline) or LBL (long baseline) mode, no angle measurements are done and only the R-group (reference) is used. Dedicated SBL/LBL transducers containing only one element or one group can be used.

14.4 Acoustic noise

14.4.1 Environmental

Noise from thrusters and propellers from surface vessels is the dominating environmental noise source. This noise is approximately 40 dB above normal sea noise. Common for all noise sources is that the noise level drops approximately 10 dB per decade with increasing frequency.

14.4.2 Noise level calculations

The noise level at the system detector is calculated by the following equation:

$$N = (N_0 - 10\log(B) - \mathrm{DI})$$

where

B = detector bandwidth,
DI = directivity of transducer.

14.4.3 Thruster noise

The noise from the thruster is changing depending on the thruster. On pitch-controlled thrusters (fixed rpm), the noise level is actually higher when running idle (0% pitch) than running with load. In addition, the impact of the thruster noise is determined by the direction of the (azimuth) thruster.

Running a thruster on low rpm and high pitch normally generates less noise than a thruster on high rpm and low pitch. In general, thrusters with variable rpm/fixed pitch generate less noise than thrusters with fixed rpm/variable pitch.

14.4.4 Sound paths

The velocity of sound is an increasing function of water temperature, pressure, and salinity. Variations of these parameters produce velocity changes, which in turn cause a sound wave to refract or change its direction of propagation. If the velocity gradient increases, the ray curvature is concave upward (Figure 14.8). If the velocity gradient is negative, the ray curvature is concave downward.

The refraction of the sound paths represents the major limitations of a reliable underwater navigation and telemetry system. The multipath conditions can vary significantly depending upon ocean depth, type of bottom, and transducer—transducer configuration and their respective beam patterns. The multipath transmissions result in a time and frequency smearing of the received signal as illustrated.

There are several ways of attacking this problem. The obvious solution is to eliminate the multiple arrivals by combining careful signal detection design with the use of a directional transducer beam. A directional receiving beam discriminates against energy outside of the arrival direction and a directional transmit beam projects the energy, so that a minimum number of propagation paths are excited.

14.4.5 Sound velocity

From Bowditch (2002), the speed of sound in seawater is a function of its density, compressibility, and, to a minor extent, the ratio of specific heat at constant pressure to that at constant volume (Figure 14.9). As these properties depend upon the temperature, salinity, and pressure (depth) of seawater, it is customary to relate the speed of sound directly to the water temperature, salinity, and pressure. An increase in any of these three properties causes an increase in the sound speed. The converse is also true.

The speed of sound changes by 3−5 m/s/°C temperature change, by about 1.3 m/s/PPT (PSU) salinity change, and by about 1.7 m/s/100 m depth change. A simplified formula adapted from Wilson's (1960) equation for the computation of the sound speed in seawater is:

$$U = 1449 + 4.6T - 0.055T^2 + 0.0003T^3 + 1.39(S - 35) + 0.017D$$

where U is the speed (m/s), T is the temperature (°C), S is the salinity (PSU), and D is depth (m).

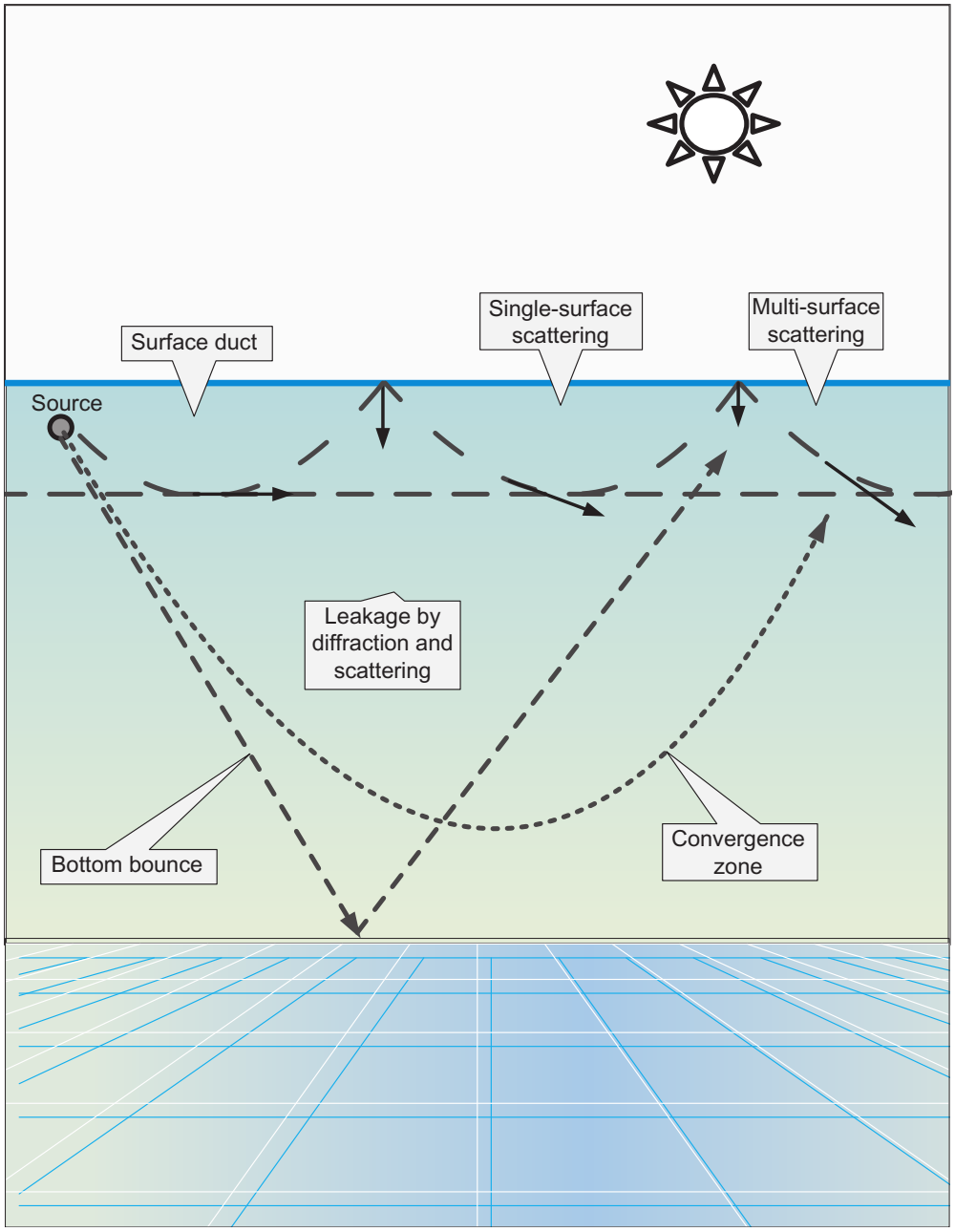


FIGURE 14.8

Ray diagrams for deepwater propagation.

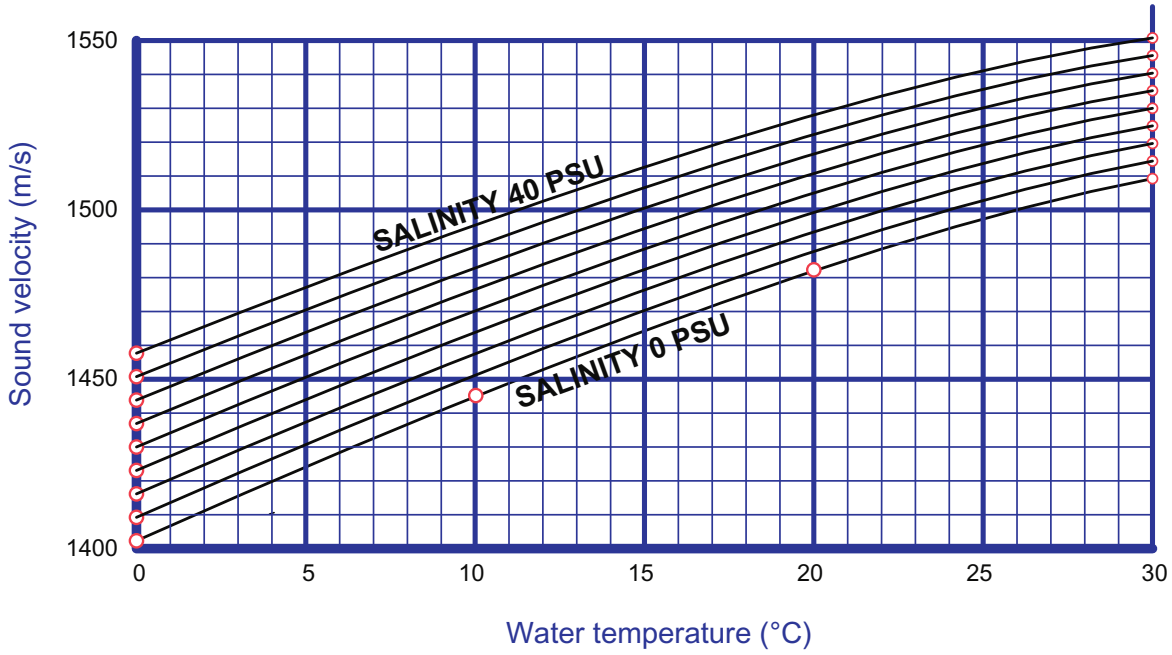


FIGURE 14.9

Example of velocity change with changing temperature and salinity.

14.4.6 Reflections

Reflections can be caused when the signal bounces off a subsea structure, seabed, riser, ship's hull, or surface (Figure 14.10). Normally the reflection is not perfect, meaning the reflected pulse has less energy than the direct pulse and should not cause problems. Sometimes the pulse is so strong it might cause problems for the pulse detection in the receiver.

When two pulses (sine waves) are added, the result can be stronger or weaker than a single pulse. Adding two signals that are 180° out of phase and of equal strength will create no signal at all (Figure 14.11).

If the signal path of the reflection is 0.5λ (or multiples of this) longer, the above situation might occur.

Example (refer to Figure 14.11):

$$f = 30 \text{ kHz} \approx \lambda = 0.05 \text{ m}$$
$$d = 100 \text{ m}$$

180° phase shift (1/2λ):

$$d_x = \sqrt{((100.025/2)^2 - (100/2)^2)} = 1.12 \text{ m}$$

180° phase shift (2 + 1/2λ):

$$d_x = \sqrt{((100.125/2)^2 - (100/2)^2)} = 2.50 \text{ m}$$

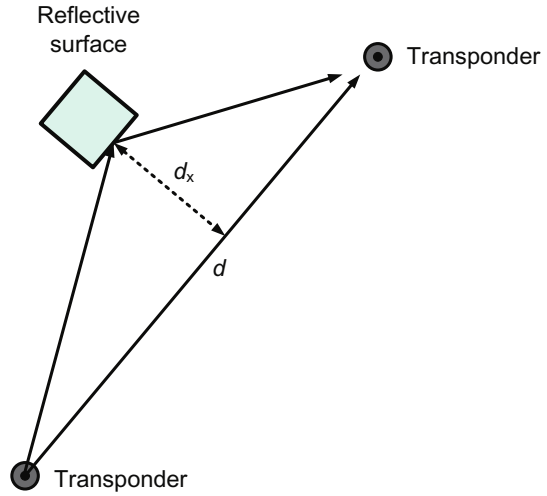


FIGURE 14.10

Two LBL transponders measuring baseline.

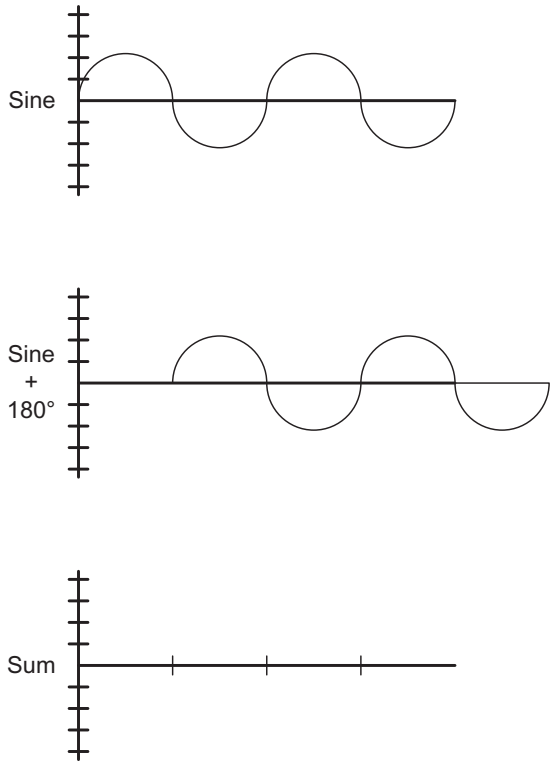


FIGURE 14.11

Sine wave interference.

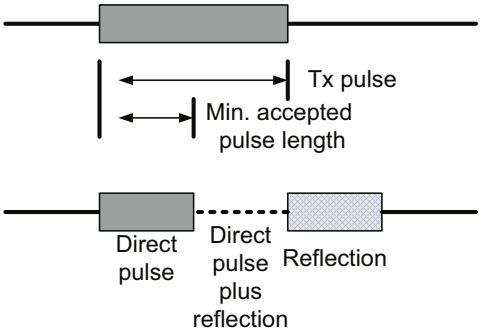


FIGURE 14.12

Received pulse.

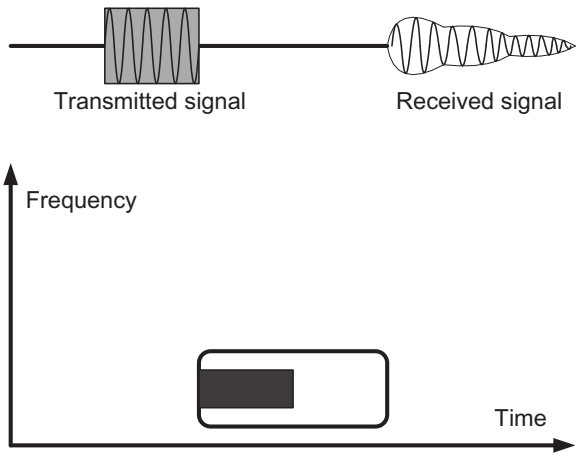


FIGURE 14.13

Multipath transmission: time and frequency smearing of acoustic signals.

This shows that a surface with 100% reflection will create no resulting signal at the receiver given the distances. The same problem can be caused by ray bending or reflections from risers or a ship's hull.

Even reflections that are slightly delayed might cause problems. The receiver has certain criteria in order to accept the signal as a pulse. One of them is the pulse length. As Figures 14.12 and 14.13 illustrate, a worst-case condition will be that the direct pulse length is too short to be acknowledged as a pulse before the reflection cancels it. Then the remainder of the reflection is just long enough to be accepted as a pulse.

CHAPTER

Sonar

15

CHAPTER CONTENTS

15.1 Sonar basics 388
15.1.1 Why sound? 388
15.1.2 Definition of sonar 388
15.1.3 Elements required for sonar equipment 389
15.1.4 Frequency and signal attenuation 389
15.1.5 Active versus passive sonar 390
15.1.6 Transducers 390
15.1.7 Active sonar 394
15.1.8 Terminology 395
15.1.9 Sonar equations 397
15.1.10 Reflectivity and gain setting 398
15.1.11 Sound backscatter 399
15.1.12 Single- versus multibeam 401
15.1.13 Frequency versus image quality 402
15.2 Sonar types and interpretation 404
15.2.1 Imaging sonar 404
15.2.2 Profiling sonar 404
15.2.3 Side-scan versus mechanically/electrically scanning 406
15.2.4 Single-/dual-/multifrequency versus tunable frequency 409
15.2.5 CHIRP technology and acoustic lens systems 410
15.2.5.1 CHIRP sonar 410
15.2.5.2 Acoustic lens sonar 413
15.3 Sonar techniques 416
15.3.1 Using an imaging sonar on an ROV 416
15.3.2 Technique for locating targets with ROV-mounted scanning sonar 418
15.3.3 Interpretation of sonar images 419
15.4 New and emerging technologies 420
15.4.1 Image capture 420
15.4.2 Image rendering 421

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In this section, the subject of sonar is presented with emphasis on "imaging sonar" systems as deployed aboard many remotely operated vehicles (ROVs). The field of sonar comprises a broad and very in-depth body of knowledge lightly dealt with in this chapter. For further reading on this subject, refer to the bibliography.

This chapter relies heavily upon contributions by experts in this field. The explanation of compressed high intensity radar pulse (CHIRP) technology comes courtesy of Maurice Fraser with Tritech International Limited. Special thanks go to Richard Marsh for arranging for its inclusion. The explanation of acoustic lens technology comes from a technical paper entitled "Object Identification with Acoustic Lenses" courtesy of Edward Belcher of Sound Metrics. And special thanks go to Willie Wilhelmsen, Jeff Patterson, and Mitch Henselwood of Imagenex Technology Corporation for their extensive contributions to this chapter over a 2-year period.

15.1 Sonar basics

15.1.1 Why sound?

Sound propagates readily through water well beyond the range of light penetration (even at light's highest penetration wavelengths). Sound propagates best through liquid and solids, less well through gases, and not at all in a vacuum. As light reflections differentiate between objects by varying levels of reflection (light intensity) as well as changing wavelengths (light color), so does sonar characterize targets through reflected sound frequency and intensity. Through proper data interpretation, target information may be discerned to identify the object through active or passive sound energy.

Some typical applications for sonar technology include:

- Echo sounding for bathymetry
- Side-scan sonar for bathymetry and item location
- Underwater vehicle-mounted imaging sonar for target identification
- Geophysical research
- Underwater communications
- Underwater telemetry
- Military listening devises (passive sonar) for submarines and shipping identification
- Position fixing with acoustic positioning systems
- Fish finding
- Acoustic seabed classification
- Underwater vehicle tracking over bottom
- Measuring waves and currents

15.1.2 Definition of sonar

Sound transmission through water has been researched since the early 1800s. The technology rapidly matured, beginning in the 1930s. With the explosion of technology in the fields of beamforming transducers and digital signal processing, today's sonar system encompasses a wide range of acoustical instruments lumped under the general heading "Sonar."

The term "sonar" is derived from "*so*und *na*vigation and *r*anging." The purpose of this technology is to determine the range and reverberation characteristics of objects based upon underwater sound propagation.

15.1.3 Elements required for sonar equipment

For sonar equipment to function, three key elements are necessary:

1. *Source*: A sound source is needed to produce the pulse energy for reflection (in active sonar systems) and/or reception (for passive sonar systems).
2. *Medium*: In the vacuum of space, sound does not travel. Some type of medium is required to transmit the sound wave energy between the source and receiver.
3. *Receiver*: Some type of receiver is needed to transform the mechanical energy (sound waves) into electrical energy (electrical signal) in order to process the sound into signals for information processing.

The source vibrates (whether it is a sonar transducer, machinery noise from an engine aboard a vessel, or the mating call of whales in the ocean) causing a series of compressions and rarefactions, thus propagating the sound through the transmission medium.

15.1.4 Frequency and signal attenuation

Beginning in 1960 and culminating with the Heard Island feasibility tests in 1991, sound propagation tests were conducted off the coast of Australia with listening stations placed around the world's oceans. What was discovered was the extreme distance low-frequency sound in seawater would travel. As the frequency increased, however, sound attenuation increased dramatically. Table 15.1 presents a general working range of various sound frequencies in seawater.

Lower frequency sound propagates through higher density materials better than higher frequency sound waves. Very low-frequency sound is used in applications such as seismic surveys of sub-bottom rock formations in search of hydrocarbons, since the sound waves will penetrate more dense strata before reflecting off of hard substrate. Low frequency is used to penetrate through mud by sub-bottom profilers. In the very high-frequency ranges, sound reflection produces only a

Table 15.1 Sonar Two-Way Working Ranges

Frequency	Wavelength	Distance
100 Hz	15 m	1000 km or more
1 kHz	1.5 m	100 km or more
10 kHz	15 cm	10 km
25 kHz	6 cm	3 km
50 kHz	3 cm	1 km
100 kHz	1.5 cm	600 m
500 kHz	3 mm	150 m
1 mHz	1.5 mm	50 m

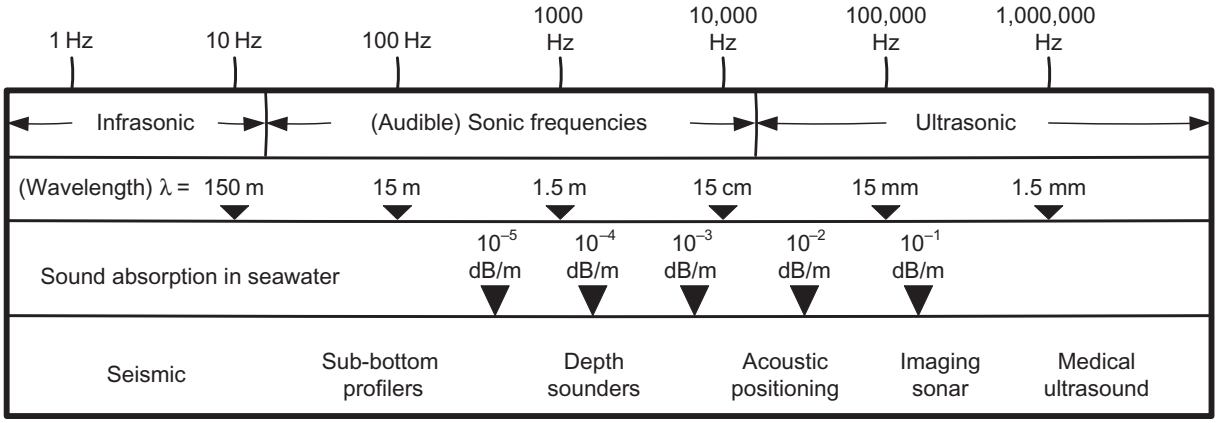


FIGURE 15.1

Acoustic spectrum.

surface paint of the target but gives higher target surface detail. An analysis of the sound frequency spectrum is provided in Figure 15.1.

15.1.5 Active versus passive sonar

Range and bearing from a sound source can be derived either actively or passively. The transducer/receiver receives sound from some source and then (through a series of computations) arrives at a range and bearing solution. *Passive* sonar essentially uses listening *only* to derive these computations. *Active* sonar (in a method similar to radar) uses a transmitter/receiver arrangement to send out an acoustic signal and then listens for a reflection (echo) of that signal back to the receiver over time to derive a range and signal strength plot.

Beam forming of sonar signals provides for various acoustic properties of the reflected signal (for active sonar), thus allowing analysis of the backscatter for target characterization. For imaging sonar, a so-called fan beam is used to depict small details, thus building a clear pictorial image of the target being insonified (Figure 15.2). For depth sounding sonar, a broad conical beam gives an indication of the closest distance to the bottom over a broader area. The typical usage of a conical beam is for depth sounding. For ROV-mounted sonar applications, a fan beam is used for the imaging sonar while a conical beam is used for the "altimeter" to determine the submersible's height above the bottom.

15.1.6 Transducers

This section will delve further into the transducer and its use in beam forming for gathering various backscatter characteristics of the target. By definition, a "transducer" is any of a number of devices that convert some other form of energy to or from electrical energy. Examples of devices that fall under the category of "transducer" are photoelectric cells, common stereo speakers, microphones, electric thermometers, any type of electronic pressure gauge, and an underwater piezoelectric transducer. This chapter will concentrate on the piezoelectric transducer and its usage with sonar

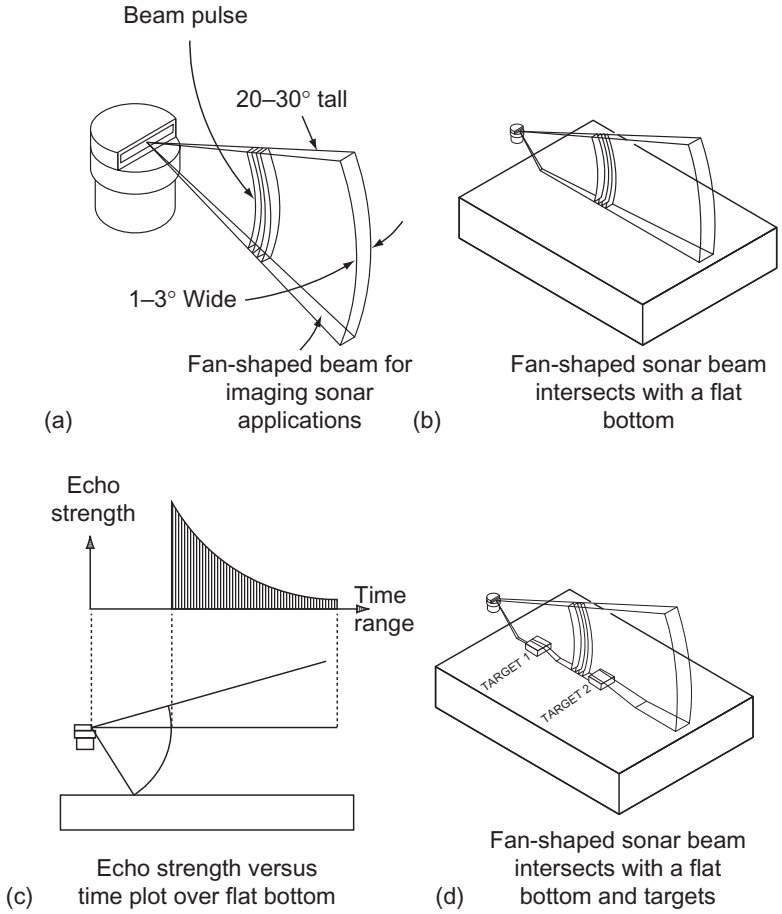


FIGURE 15.2

Single beam sonar (a) graphically defined, (b) pulsed over bottom through water column, (c) echo strength plotted over time, and (d) depicted with sample targets.

systems. This analysis begins with the familiar land-based transducer, the home stereo speaker, and then progresses into the unfamiliar underwater transducer.

In a stereo speaker, any number of technological twists is used to convert electrical energy to acoustic energy. The most basic means involves the varying of a voltage within a coil wrapped around an electromagnet moving a paper cone. The cone vibration produces acoustic energy in the form of sound waves propagating in an omnidirectional fashion as depicted in Figure 15.3. As shown in the figure, the sound moves through the medium (in this case air) through a series of compressions and rarefactions as the wave travels at sonic velocity away from the sound source. As the cone vibrates, it produces sound in a bidirectional pattern. In order to change the sound pattern from an omnidirectional wave pattern to a more focused sound projection, enclosures and sound projectors are used to focus the sound beam to a desired direction. If a home stereo speaker is disassembled, one would find three basic speakers (the woofer, midrange, and tweeter), each with frequency response based upon its size and nominal resonance frequency.

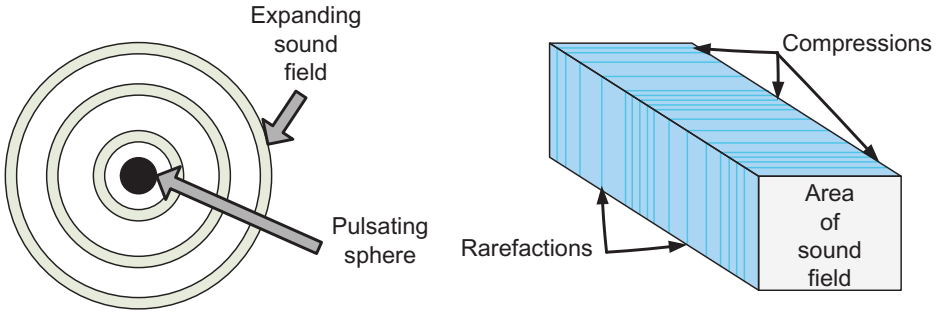


FIGURE 15.3

Sound pulse traveling in wave fronts through a medium.

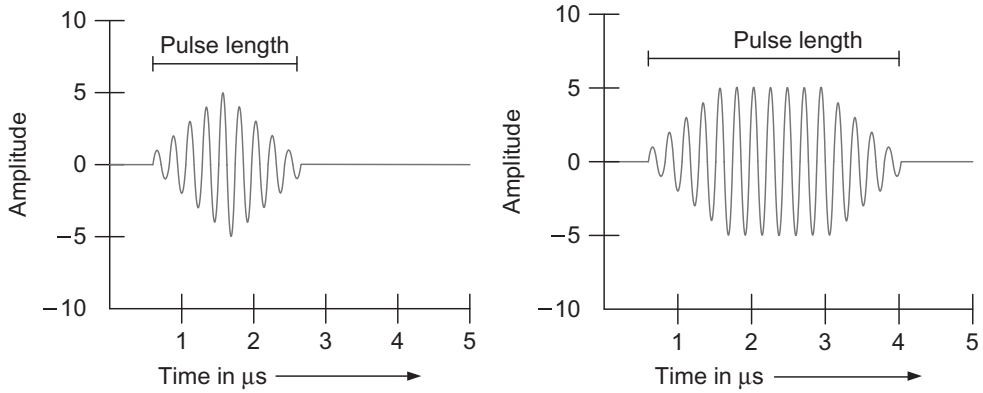


FIGURE 15.4

Pulse length as a function of time.

The most prevalent underwater transducer is the polycrystalline, "piezoelectric" material PZT (lead zirconium titanate), which was discovered in the 1940s to be an effective transducer. As discussed in Chapter 14, piezoelectric underwater transducers emit acoustic pulses through vibration when an electric signal is sent through the silver deposits at the poles of the ceramic core. Piezoelectric underwater transducers are resonant; therefore, a transducer is sized to match a certain nominal narrowband frequency in which it is most sensitive. The sonar system is then designed around this nominal frequency.

On a typical active transducer, the transducer will vibrate at the frequency of the electrical signal being applied, not necessarily the resonant frequency. If that frequency is near the resonant frequency, the transducer is freer to respond, giving a larger amplitude signal than if the frequency is far from the resonant frequency. The length of time the transducer is activated determines the pulse length generated (Figures 15.4 and 15.5). A short pulse length allows better discrimination between targets, but may not allow enough acoustic energy to reflect off targets as distance increases.

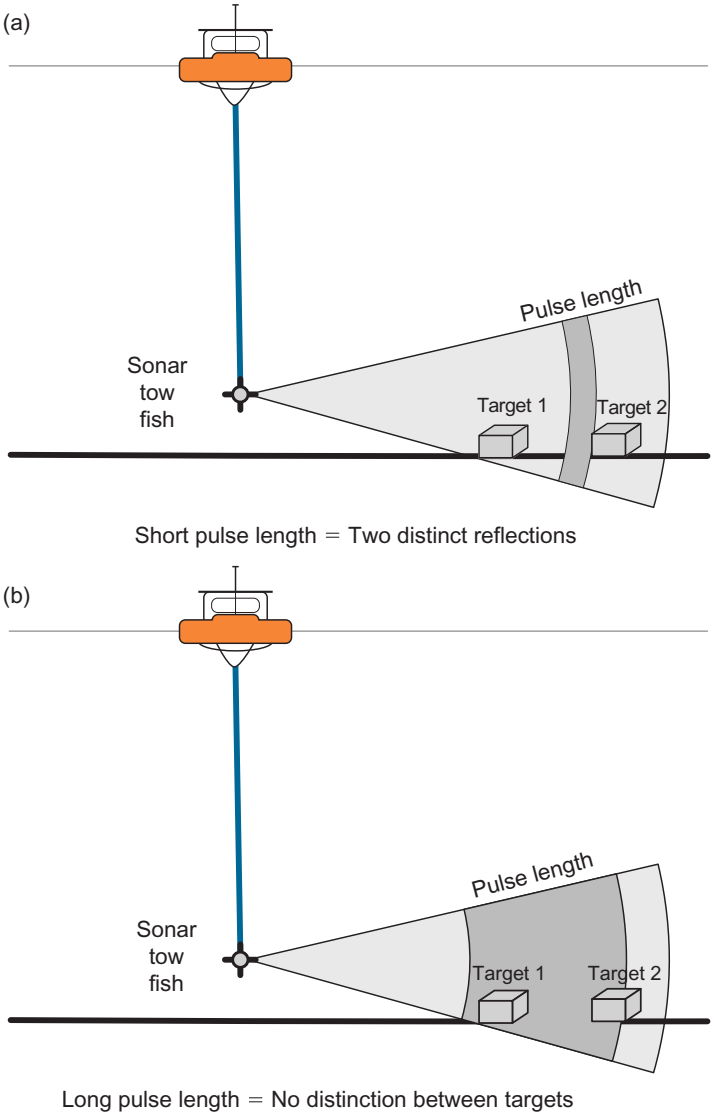


FIGURE 15.5

Pulse length for discrimination between targets with (a) short pulse length and (b) longer pulse length.

The directivity index for a transducer defines the ratio between an omnidirectional point source, that is, a circular (two-dimensional, 2D) or spherical (three-dimensional, 3D) sound source with no directivity, to the source level intensity on the beam axis of a directional sound pulse. The reason this factor is important is that the beam form is defined by its directivity index as well as its vector (Figure 15.6).

As the directivity index of a sound beam becomes more focused, some side effects become pronounced, thus requiring consideration. The main beam of a directional sound source is called the "main lobe" and is the primary pulse directed toward the target. As depicted in Figures 15.7 and 15.8,

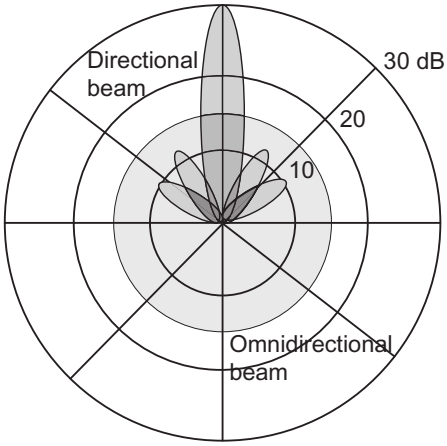


FIGURE 15.6

Directivity index for a circular as well as a directional sound beam.

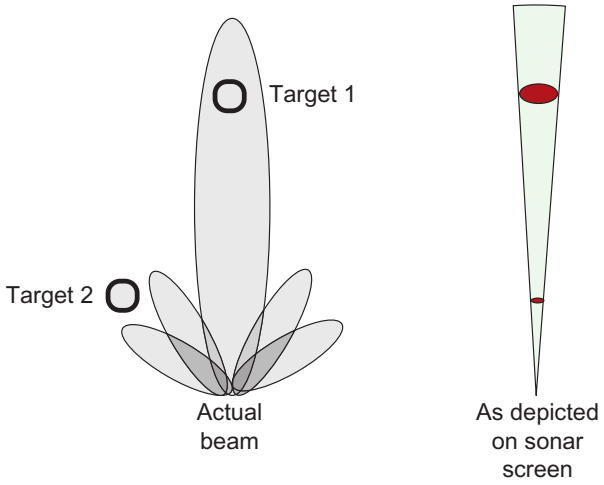


FIGURE 15.7

Target in the side lobe of this beam will produce an echo at the receiver.

a by-product of this main lobe, side lobes, develop that are also subject to backscatter off secondary targets. An analogy to this concept is the simple push of the hand on the surface of a swimming pool, directing a wave pulse in a specific direction. As the hand is pushed in one direction, it can be noticed that the main pulse pushes in the direction of the wave but side waves emanate on either side of the main wave.

15.1.7 Active sonar

The majority of civilian sonar systems falls under the classification of active sonar. Active sonar systems make use of beam forming, frequency shifting, and backscatter analysis to characterize

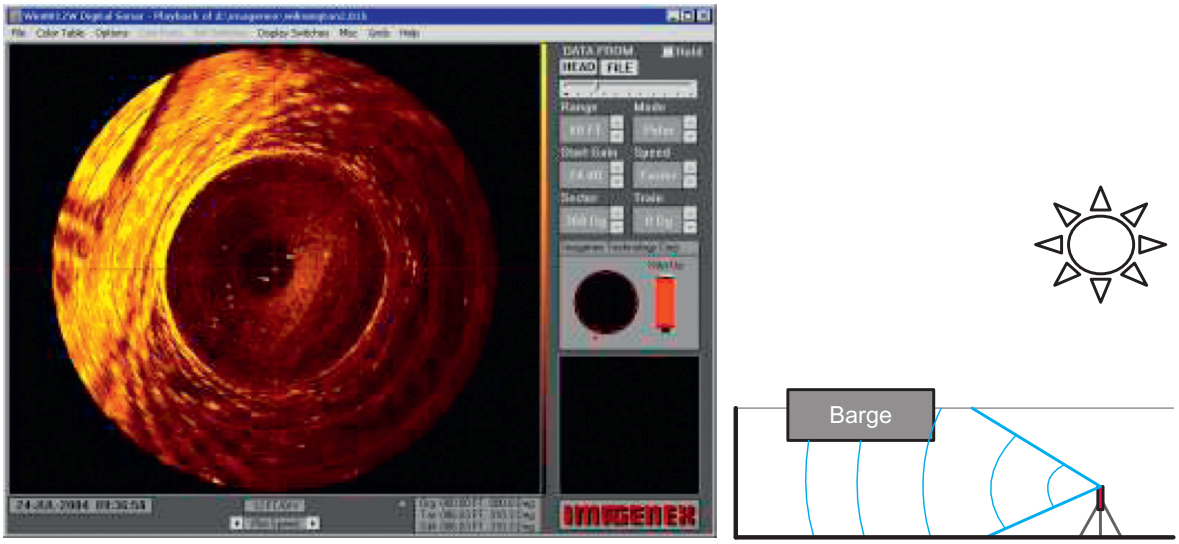


FIGURE 15.8

Mechanically scanning sonar viewing barge with scalloped lines from side lobe backscatter.

targets insonified. There are a vast number and category of active sonar products on the market to fit various applications (with the technology advancing daily). Any number of these technologies is available for mounting on ROV equipment for delivery to the work site.

15.1.8 Terminology

The following provides an explanation of the basic terms used to describe sonar techniques:

- *Angle of incidence*: The angle at which a sound reflects upon a surface (in degrees). Example: If a flashlight is shone on a mirror, the light will reflect directly back to the flashlight if there is a zero angle of incidence, reflected 90° with a 45° angle of incidence, etc. The same principle applies to a sound reflector.
- *Beam forming*: The principle of forming the sound propagation wave to provide the desired data from the wave front.
- *Color*: The different colors used to represent the varying echo return strengths.
- *Echo*: The reflected sound wave.
- *Echo return*: The time required for the echo to return to the source of the sound.
- *Pulse width*: The width of a sound propagation wave (in feet, meters, or microseconds) generated from some source. This factor has an influence on the quality of the image generated by the sonar receiver.
- *Target*: The object requiring characterization.
- *Target strength (TS)*: A measure of the reflectivity of the target to an active sonar signal.
- *Reflectivity*: The ability of a material to reflect sound energy.

- *Transmission (sound) loss (TL)*: Total of all sound losses incurred between the sound source and the ultimate receiver. Transmission losses come in two main types: spreading loss and attenuation loss.
- *Spreading loss*: Sound energy loss due to geometrical spreading of the wave over an increasingly large area as the sound propagates. Spreading losses are considered on either a 2D cylindrical plane (horizontal radiation only, or thermal layer, or large ranges compared to depth) or a 3D spherical profile (omnidirectional point source).
- *Attenuation loss*: Generally considered to be the lumped together sum of losses produced through absorption, spreading, scattering, reflection, and refraction.
- *Absorption loss*: The process whereby acoustic energy is absorbed by a material, thus producing heat. Absorption loss increases with higher frequency.
- *Scattering and reverberation*: Sound energy bouncing or reflecting off items or surfaces.
- *Noise level (NL)*: Total noise from all sources potentially interfering with the sound source reception. Therefore, Noise level (*NL*) = Self-noise (*SN*) + Ambient noise (*AN*).
- *Self-noise (SN)*: Noise generated from the sonar reception platform potentially interfering with reception of the sound source. Examples of this are machinery noise (pumps, reduction gears, power plant, etc.) and flow noise (high hull speed, hull fouling, such as barnacles or other animal life attached to hull), and propeller cavitations.
- *Ambient noise (AN)*: Background noise in the medium potentially causing interference with signal reception. Ambient noise can be either hydrodynamic (caused by the movement of water, such as tides, current, storms, wind, rain, etc.), seismic (i.e., movement of the earth—earthquakes), biological (i.e., produced by marine life), or by ongoing ocean traffic (i.e., noise caused by shipping).
- *Source level (SL)*: Sound energy level of sound source upon transmission.
- *Directivity index (DI)*: The ratio of the logarithmic relationship between the intensity of the acoustic beam versus the intensity of an omnidirectional source.
- *Signal-to-noise ratio (SNR)*: The ratio to the received echo from the target to the noise produced by everything else.
- *Reverberation level (RL)*: The slowly decaying portion of the backscattered sound from the sound source.
- *Detection threshold (DT)*: The minimum level of received signal intensity required for an experienced operator or automated receiver system to detect a target signal 50% of the time.
- *Figure of merit (FOM)*: The maximum allowable one-way transmission loss in passive sonar and the maximum two-way transmission loss in active for a detection probability of 50%. These are termed "active" (*AFOM*) and "passive" (*PFOM*). This concept is especially pertinent to the anti-submarine warfare community.
- *Shadows*: The so-called shadowgraph effect is an area of no sonar reflectivity due to blockage. The best analogy for this would be to shine a light onto an object a few feet away. The object will block the light directly behind it and cast a shadow. A shadow is depicted on a sonar display by an area behind a blocking mechanism where there is no sonar reflection.
- *Sonar*: The principle used to measure the distance between a source and a reflector (target) based on the echo return time.
- *Sound losses (absorption, spreading, scattering, attenuation, reflection, and refraction)*: Sound energy loss through various factors influencing the reception and display of reflected sound waves.

15.1.9 Sonar equations

Sonar equations look at signal losses compared to signal sources to determine the likelihood of detection of the sound source. For the passive sonar, only one-way sound transmission and noise levels are considered in deriving the minimum detection threshold, whereas on active sonar systems the sound must travel first from the sound source (the transducer of the active sonar system) through the water column to the target and then (after the backscatter) back to the receiver (Figure 15.9).

The passive sonar equation:

$$SL - TL - NL + DI \geq DT$$

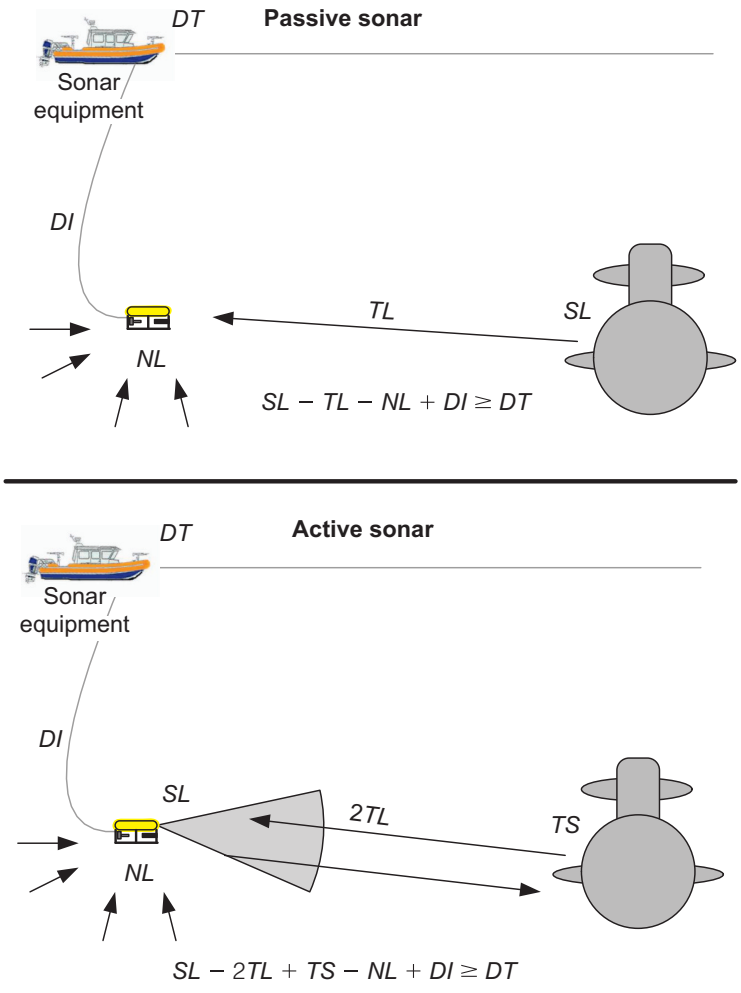


FIGURE 15.9

Sonar equations for (a) passive and (b) active sonar.

The active sonar equations:
Ambient noise limited:

$$SL - 2TL + TS - NL + DI \geq DT$$

Reverberation noise limited (reverb > ambient noise):

$$RL > NL + DI$$
$$SL - 2TL + TS - RL \geq DT$$

The figure of merit equations:

$$PFOM = SL - NL + DI - DT$$
$$AFOM = SL + TS - NL + DI - DT$$

15.1.10 Reflectivity and gain setting

Different materials reflect sound with different efficiency (Table 15.2). Mud will reflect sound very poorly while water will not reflect sound at all. The closer the substance's consistency is to water, the lower the reflectivity index. Therefore, a very good feel for the target's makeup can be gained simply by the target's level of reflectivity.

The gain setting on the sonar system will allow the operator to pick up detail within the reflection. If the gain setting is set high while surveying a sandy bottom, the screen will display no contrast between targets, since everything will show a high reflectivity value. Likewise, if the gain is set too low with a mud bottom, no detail will display, since practically all of the reflections from the bottom will be below the display setting and will be rejected.

In Figure 15.10, differing compositions on this combination of sand and mud bottom allow for discrimination between bottom makeup as well as targets standing proud of the bottom with differing makeup. The bottom of this figure displays a practice course as well as two targets easily discernible from the mud bottom. This figure depicts differing bottom composition through varying echo strength with a fixed gain setting.

In Figure 15.11, an example of a swimming pool is used to illustrate the effect of varying angles of incidence on sonar reflectivity. The tripod-mounted sonar was placed at the center of a varying shaped pool (Figure 15.11c). The walls of the pool have a very high reflectivity at the zero incidence point with much lower backscatter on the higher angles of incidence.

As a further example of incidence versus reflectivity, consider a sonar source placed in an underwater room (Figure 15.12). The sonar will insonify the room with the highest target strength

Table 15.2 Sample Reflectivity Indexes

Substance	Reflectivity
Mud	Low
Sand	Medium
Rock	High
Air/air-filled	Very high

coming from the zero angle of incidence along with another "bright spot" located in the corners. The corners will not be depicted as square due to the sound multipath as the beam approaches the corner point. Instead of a square corner, the sonar display will depict a rounded corner due to the sound reflection as the beam sweeps the corner.

15.1.11 Sound backscatter

Backscatter is the reflected sound from any object being insonified—the analysis of which is the subject of active sonar systems. Backscatter is analyzed for any number of parameters to solve particular operational needs. Some examples of backscatter analysis follow:

- Multiple bounce depth sounder backscatter is analyzed with acoustic seabed classification systems to determine the texture and makeup of the sea bottom (sand, mud, rock, oyster bed, kelp, etc.) for environmental monitoring as well as vessel navigation.
- Doppler shift backscatter is used for vehicle speed over ground as well as current/wave profiling.
- Frequency shift backscatter is analyzed in CHIRP sonar systems (see Figures 15.22–15.24) to discriminate between objects in close proximity.
- Simple high-frequency backscatter can characterize a target as to aspect, texture, surface features, and orientation.

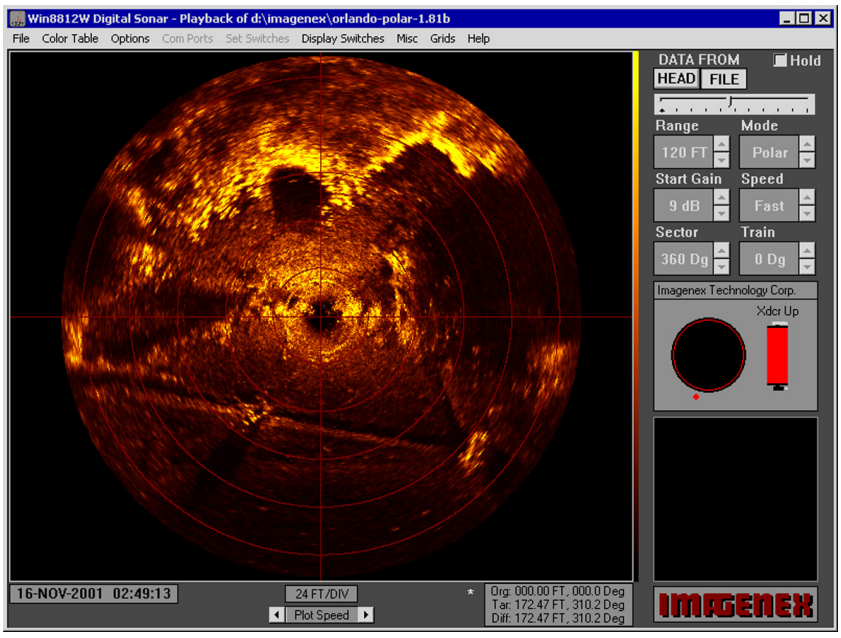


FIGURE 15.10

Combination of sand and mud bottom showing differing reflectivity based upon bottom composition (gain setting 9 dB).

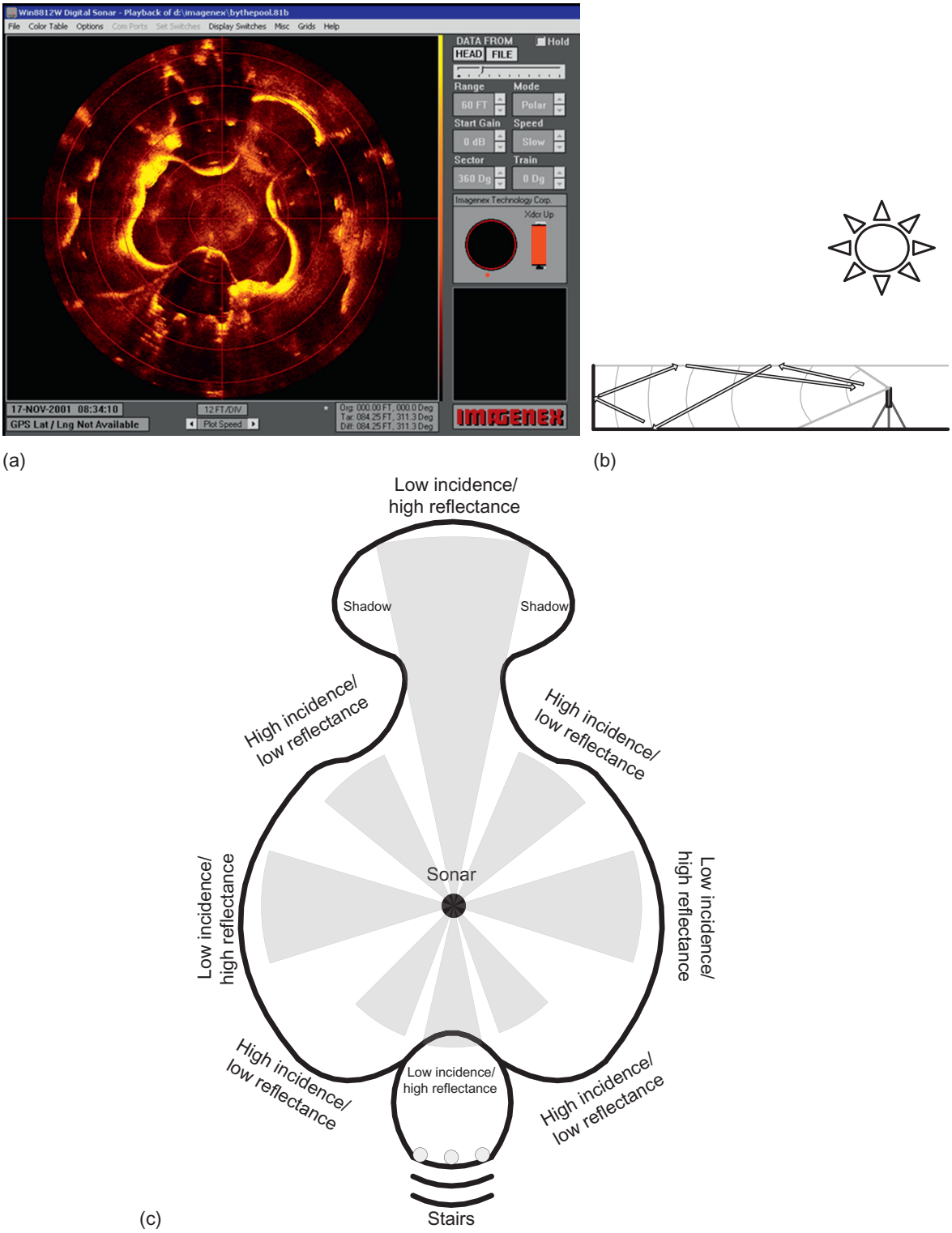


FIGURE 15.11

(a) Sonar display of a recreational swimming pool plus (b) profile view and (c) plan view containing incidence/reflectance levels per pool quadrant.

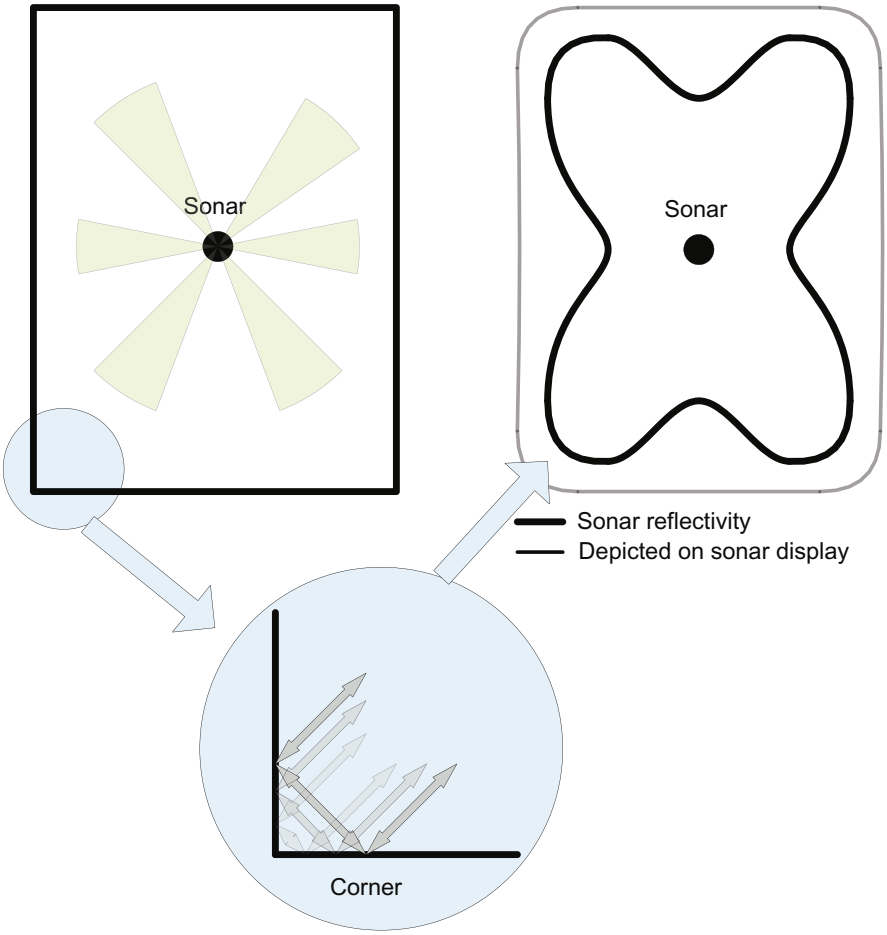


FIGURE 15.12

Sonar picture of an underwater rectangular room.

15.1.12 Single- versus multibeam

Active imaging or profiling sonar systems generally fall into three basic categories:

1. Multibeam
2. Mechanically or side-scanning sonar (single beam)
3. Single-beam directional sonar

Single-beam sonars are simply one pulse with one reception on a single receiving element. The single-beam mechanically scanning system comprises the lion's share of ROV-mounted sonar systems currently on the market today due to their simplicity as well as their reduced cost.

Multibeam sonar systems, on the other hand, transmit one wide pulse and receive the backscatter on a large number of receiving elements. Accurate sensing of the delays in sound arrival between elements enables the sonar to distinguish the direction of the received sound, in order to build up a detailed profile of the area being insonified.

In its simplest form, multibeam sonars operate on the principle of additive sound pulses. Recall from Figure 15.3 that sound spreads in an isotropic fashion (the same in all directions) from a sound projector known as an "isotropic source" (a sound projector projecting a sound source equally in all directions). Placing two isotropic sound sources in proximity proportional to the projector's wavelength, the additive pulses (termed "interference") can be either constructive or destructive based upon the phase of the pulses (Figure 15.13).

From this constructive interference, a highly directional pulse can be constructed along the line of equidistance with spacing $\lambda/2$. This forms the axis of a pulse in a 3D directional beam (Figure 15.14). If several projectors are aligned in a row, the beam can become more focused and highly directional.

By varying the time delay between the projector transmissions, the axis of the beam can be steered to project multiple beams. With the use of the same concept upon reception, the time variance can also be analyzed to accept sound sources only from a narrowly directive reception cone, thus forming the multiple beam transmission and reception. The reception array is perpendicular to the projector array so that reception is only allowed at the crossing points, allowing for clean multibeam transmission and reception.

Table 15.3 analyzes some of the benefits of multibeam technology. Multibeam sonar systems are an amazing recent technological development brought on by advances in digital signal processing of sonar returns as well as advances in highly directional transducer receiving elements. With the technology trickling down to the smaller ROV system through lower size costs and increased functionality, look for more multibeam sonar systems aboard these vehicles.

15.1.13 Frequency versus image quality

Determination of the optimum sonar frequency for the particular application at hand is a trade-off of desired image quality, depth of penetration into the medium as well as range of desired coverage area. As discussed above, the higher the frequency of the source, the higher the attenuation of the signal, thus lowering the effective range of the beam. The lower the frequency, the longer the

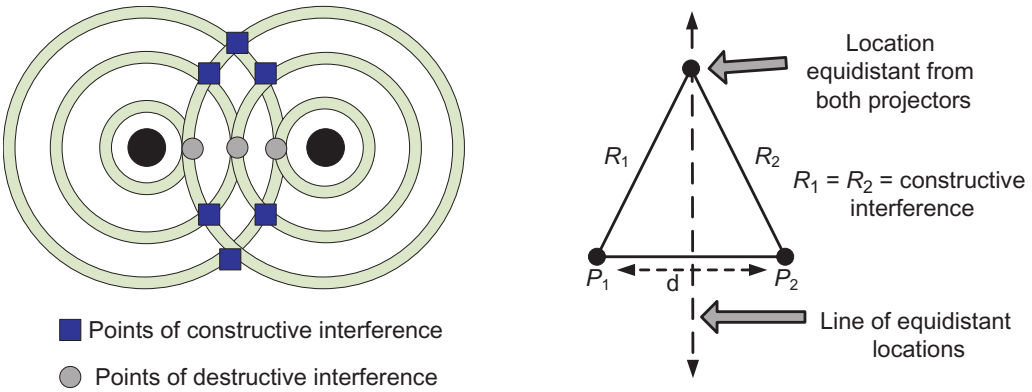


FIGURE 15.13

Constructive and destructive interference of two point sources with spacing *d*.

propagation distance and the penetration depth into the medium being insonified. The higher the frequency, the lesser the penetration into the medium, allowing for a more detailed surface paint of the target for target shape and texture classification.

For small-item surface discrimination, the wavelength of the pulse must be no larger than the target—and much shorter for any detailed imaging of the target. For better small-object discrimination, the higher the frequency the better. What suffers at very high frequencies

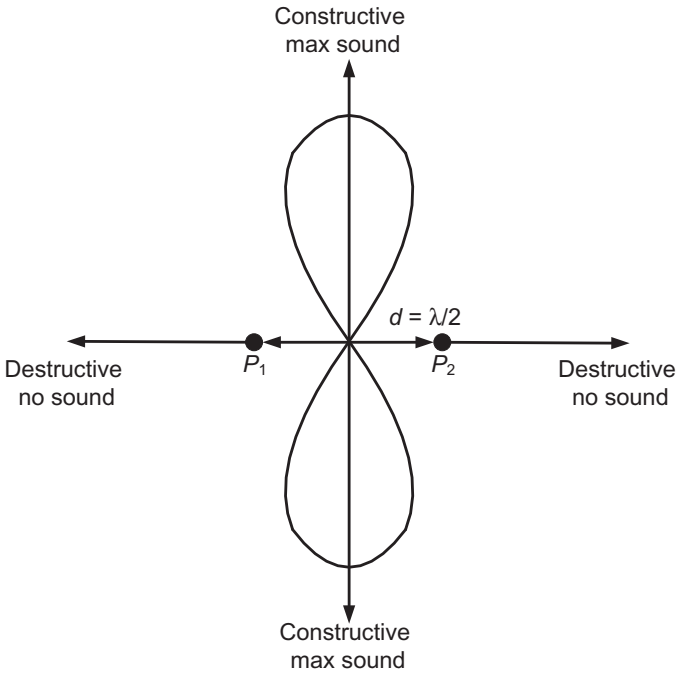


FIGURE 15.14

Constructive dipole projection of sound along a beam axis.

Table 15.3 Some Advantages and Disadvantages of ROV-Mounted Single and Multibeam Sonar Systems

Sonar System	Advantages	Disadvantages
Single beam	• Able to view 360° around vehicle	• Slow image generation
Multibeam	• Up to 10 frames per second image update rate at short range	• ROV-mounted projector allows narrow reception beam width • Higher computing power needed drives cost upward

(700 kHz and higher) is the useful range of the sonar system. For searching a wide area for small-object location, a range of only 20 m will require an extended search time to cover any appreciable area. On a more human note, medical ultrasound functions in the 2−3 MHz range (and higher) for very-small-object discrimination. At these frequencies, extremely high detail can be achieved over very short distances.

Low-frequency sonar is used for such applications as seismic surveys of geological strata (very low frequency—in the 60−100 Hz range) as well as sub-bottom profiling below the mud line (low frequency—in the 10−100 kHz range, depending upon desired mud penetration depth). These systems can gain a broad classification of the overall area, but small details of individual areas are not possible due to the wavelength being larger than any small target.

15.2 Sonar types and interpretation

15.2.1 Imaging sonar

A fan-shaped sonar beam scans surfaces at shallow angles, usually through an angle in the horizontal plane, and then displays color images or pictures. The complete echo strength information for each point is displayed primarily for visual interpretation. As depicted in Figure 15.15, with imaging sonar a fan-shaped sonar beam scans a given area by either rotating (as with ROV-mounted systems) or moving in a straight line (as with side-scan sonar systems).

A pulse of sound traveling through the water generates a backscatter intensity (or amplitude) that varies with time and is digitized to produce a time series of points. The points are assigned a color or grayscale intensity. The different colored points, representing the time (or slant range) of each echo return, plot a line on a video display screen. The image (Figure 15.16), consisting of the different colored points or pixels, depicts the various echo return strengths.

The following characteristics are necessary to produce a visual or video image of the sonar image:

- The angle through which the beam is moved is small.
- The fan-shaped beam has a narrow angle.
- The transmitted pulse is short.
- The echo return information is accurately treated.

These visual images provide the viewer with enough data to draw conclusions about the environment being scanned. The operator should be able to recognize sizes, shapes, and surface-reflecting characteristics of the chosen target. The primary purpose of the imaging sonar is to act as a viewing tool.

15.2.2 Profiling sonar

Narrow pencil-shaped sonar beams scan surfaces at a steep angle (usually on a vertical plane). The echo is then displayed as individual points or lines accurately depicting cross-sections of a surface. Echo strength for each point, higher than a set threshold, digitizes a data set for interfacing with external devices.

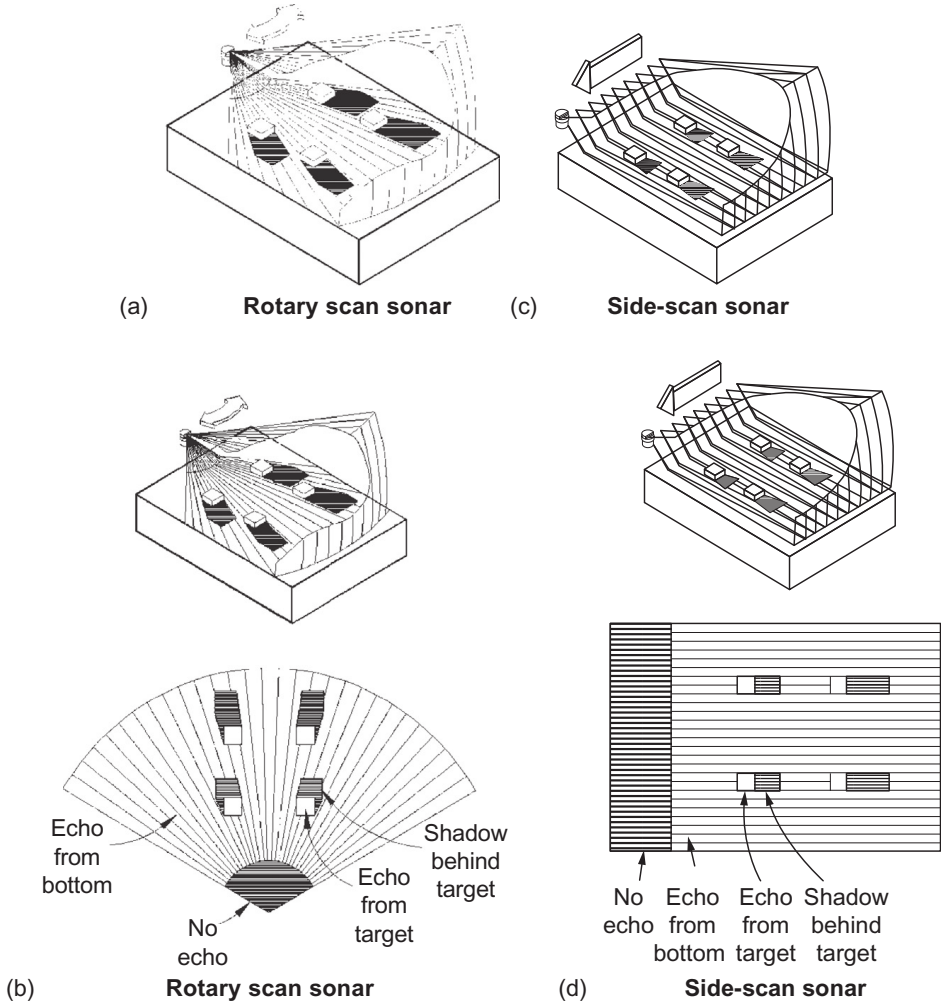


FIGURE 15.15

An imaging sonar builds a picture (a) and (b) via rotation of the head (i.e., mechanically scanning) or (c) and (d) with motion through the water (i.e., side scan).

The data set is small enough to be manipulated by computer (primarily a measurement tool). In profiling, a narrow pencil-shaped sonar beam scans across the surface of a given area, generating a single profile line on the display monitor (Figure 15.17). This line, consisting of a few thousand points, accurately describes the cross-section of the targeted area.

A key to the profiling process is the selection of the echo returns for plotting. The sonar selects the echo returns, typically one or two returns for each "shot," based on a given criterion for the echo return strength and the minimum profiling range. The information gathered from the selection criteria forms a data set containing the range and bearing figures. An external device, such as a

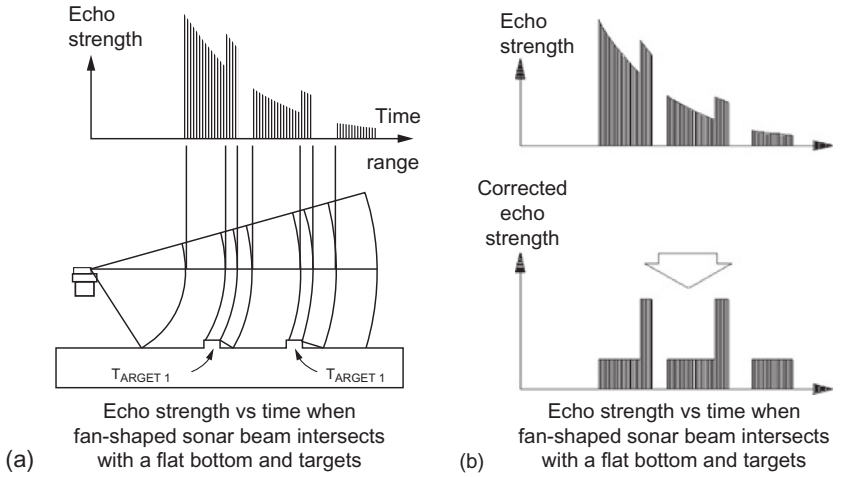


FIGURE 15.16

Echo strength versus time and range (a) plotted in raw form and (b) normalized over full range.

personal computer or data logger, accesses the data set through a serial (or other communications protocol) interface with the sonar.

The profile data is useful for making plots of bottom profiles, trench profiles, and internal and external pipeline profiles. The primary purpose of the profiling sonar is as a quantitative measuring tool such as a depth sounder or for bottom characterization. Imaging sonar can also be combined with profiling sonar to accomplish both imaging and altitude measurement (distance from bottom) on the same platform (Figure 15.18).

15.2.3 Side-scan versus mechanically/electrically scanning

The only difference between the side-scan sonar and the mechanically/electrically scanning sonar systems is the means of locomotion of the transducer head (Figure 15.19).

The side-scan sonar generates an image of the target area by the simple means of "take a shot," move the tow fish, "take a shot," move the tow fish, etc. A highly directive pulse of acoustic energy is transmitted/bounced/received as the transducer platform is either towed behind a moving platform (surface or submerged vessel) or mounted to the side of a self-propelled platform (i.e., an AUV). Problems encountered with image generation by a side-scan sonar tow fish involve the constant surge and stall of the tow fish as the tow platform (e.g., a boat) bounces and pitches over a heavy sea state. AUVs (with vehicle-mounted side-scan sonars) are perceived to be potentially superior as an image generation platform due to the decoupling of the sonar platform from the surface sea state, allowing for steady and smooth image generation without image smear.

As depicted in Figure 15.15, the mechanically or electronically scanning sonar is mounted to some other relatively stable platform (ROV, tripod, tool, dock, etc.). For the mechanically scanning sonar system, the transducer head is rotated via a stepper motor timed to move as the transmit/receive cycle completes for that shot line. An image is generated as the sonar head builds the sonar lines in either a polar fashion or in a sector scan (Figure 15.20).

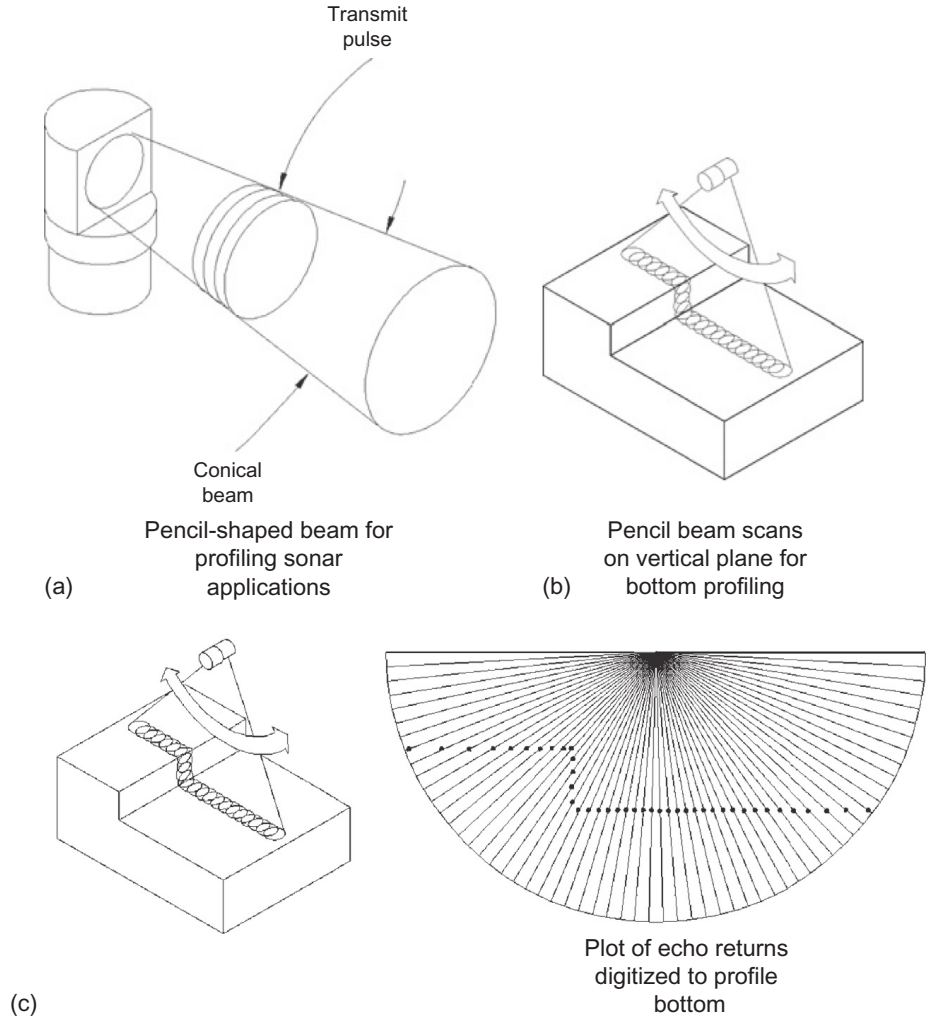


FIGURE 15.17

A profiling sonar (a) beam depicted, (b) beam scanning of bottom, and (c) display generation of beam scan.

The electrically scanning sonar can either be a spaced steerable array, a multibeam sonar, or a focused array:

- A steerable array involves separate point sources timed to produce a dipole beam. The time spacing of the pulses allows the beam axis to be steered due to additive pulse fronts.
- The multibeam sonar system is normally fixed with a nominal beam width of no more than 180°. The single sonar pulse is generated with multiple high directivity index receivers discriminating the backscatter return over its entire beam width to produce an image of the area under investigation.
- The focused array and acoustic lens sonar technologies are both incredible technologies that focus the acoustic beams onto a localized point at very high frequencies, generating near picture quality images with a high frame rate. More on this technology is detailed at Section 15.2.5.2 below.

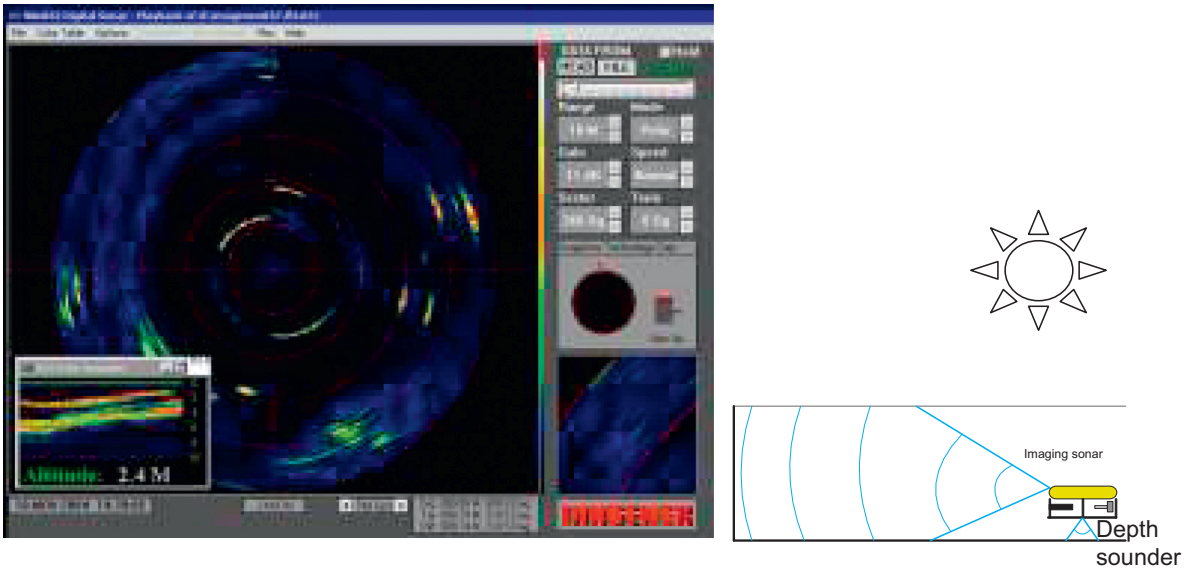


FIGURE 15.18

Here is an example of a dual imaging sonar and altimeter combination (image on screen and altitude on left).

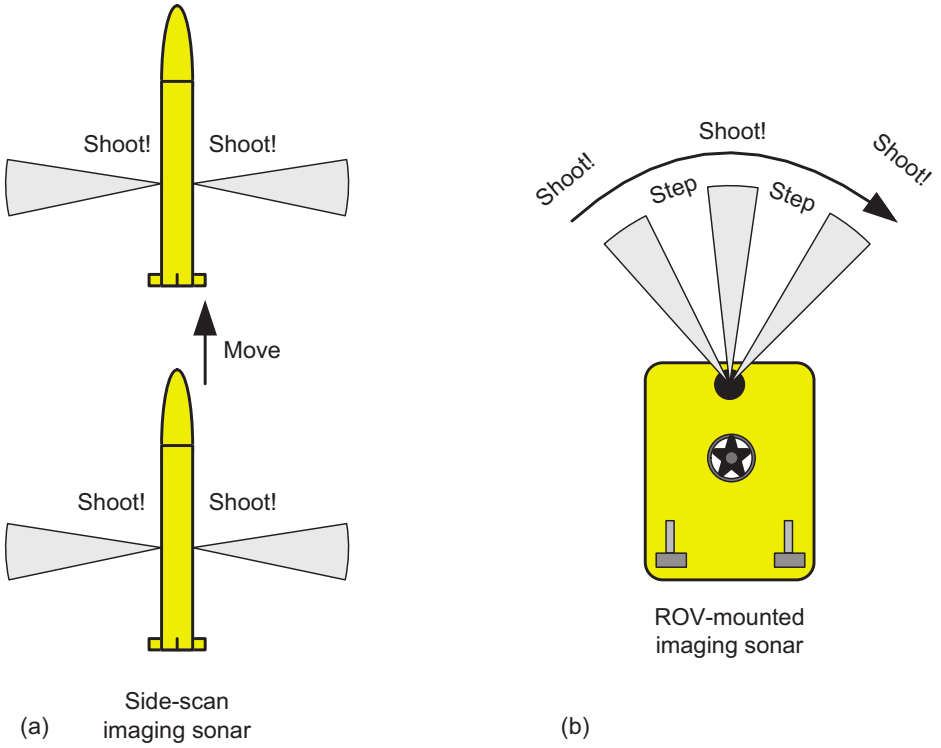


FIGURE 15.19

(a) Side-scan sonar versus (b) ROV-mounted mechanically scanning sonar.

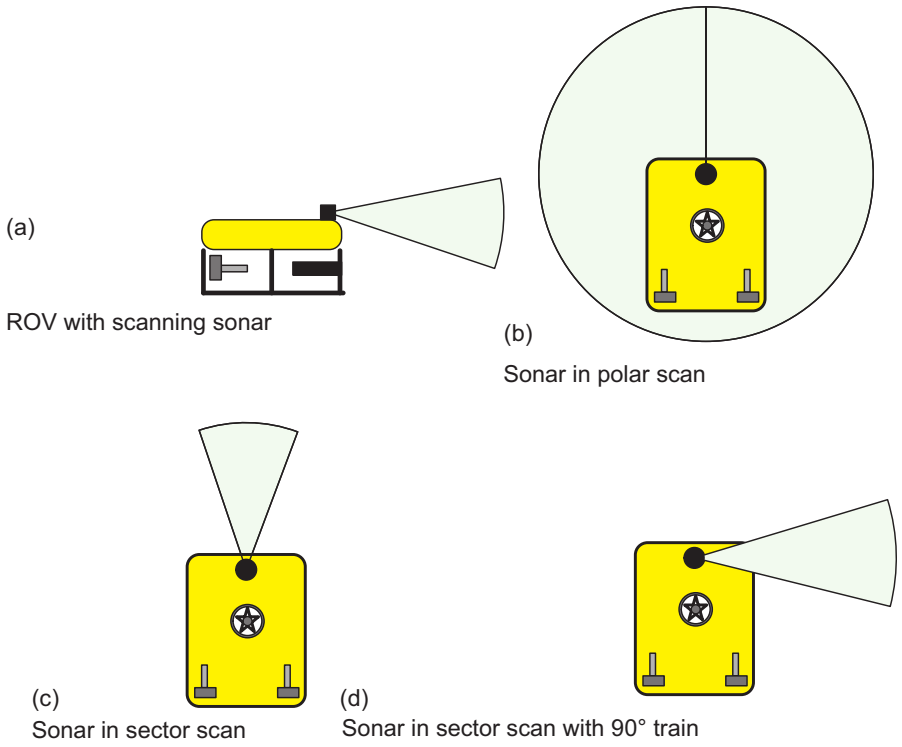


FIGURE 15.20

Scan modes of a mechanically scanning sonar system with (a) profile view, (b) plan view of polar mode, (c) sector scan, and (d) sector scan with train.

15.2.4 Single-/dual-/multifrequency versus tunable frequency

The single-/dual-/multifrequency, CHIRP, and tunable sonars all have their advantages and disadvantages. Definitions and discussion for each follows:

1. *Single frequency*: The single-frequency sonar system transmits and receives on one frequency and is the simplest sonar design due to its transducer selection on its nominal resonance frequency. Unfortunately, only a limited amount of target data can be ascertained from the use of a single-frequency backscatter analysis.
2. *Dual frequency*: This system arrangement allows gleaning of differing data parameters from the same target area based upon simultaneous or alternating transmissions of dual-frequency acoustic beams, generating each frequency's backscatter characteristics of the sample area. An excellent example of the advantages of dual-frequency sonar systems came recently during an oil spill in the Gulf of Mexico. A barge filled with heavy (heavier than water) heating oil struck a submerged unmarked platform destroyed by one of the storms of 2005. The oil sank to the bottom, requiring various governmental agencies to track the plume. To discriminate the submerged oil plume from the loosely consolidated mud bottom, a dual-frequency

(100 kHz/500 kHz) towed Klein 5000 sonar was used. The low-frequency band penetrated through the relatively light viscosity oil while the higher frequency bounced off the top of the oil/water interface, allowing for differentiation of the relatively rough mud bottom from the smooth oil as it migrated on the bottom (Figure 15.21). Once the search area was covered and the sonar data analyzed, an observation-class ROV visually characterized the bottom based upon sonar returns. Excellent tracking results were obtained of the oil plume, allowing for a higher incidence of recovery.

3. *CHIRP sonar*: This frequency-shifting sonar technology (along with its numerous technological advantages) is described more fully in the next section.
4. *Tunable frequency*: This type of transmitter/receiver combination uses frequency tuning to glean differing characteristics of the target area based upon the backscatter characteristics of the frequency transmitted. The strength of this technique is the ability to insonify the target and analyze backscatter at the various frequencies to gain better target characterization. The weakness is the rapid degradation in efficiency of the transducer as the frequency departs the nominal transducer design range.

15.2.5 CHIRP technology and acoustic lens systems

15.2.5.1 CHIRP sonar

CHIRP (compressed high intensity radar pulse) techniques have been used for a number of years above the water in many commercial and military radar systems. The techniques used to create an electromagnetic CHIRP pulse have now been modified and adapted to commercial acoustic imaging sonar systems.

To understand the benefits of using CHIRP acoustic techniques, one needs to analyze the limitations using conventional monotonic techniques. An acoustic pulse consists of an on/off switch modulating the amplitude of a single carrier frequency (Figure 15.22).



FIGURE 15.21

Dual-frequency sonar used during oil spill plume tracking with (a) tow fish hand launch and (b) sonar display.

The ability of the acoustic system to resolve targets is determined by the pulse length; this, however, has its drawbacks. To get enough acoustic energy into the water for good target identification and over a wide variety of ranges, the transmission pulse length has to be relatively long. The equation for determining the range resolution of a conventional monotonic acoustic system is given by:

$$Range\ resolution = velocity\ of\ sound/(bandwidth \times 2)$$

In a conventional monotonic system at moderate range, a typical pulse length is 50 μs and velocity of sound (VOS) in water 1500 m/s (typical). Therefore, the range resolution = 37.5 mm. This result effectively determines the range resolution (or ability to resolve separate targets) of the monotonic acoustic imaging system (Figure 15.23).

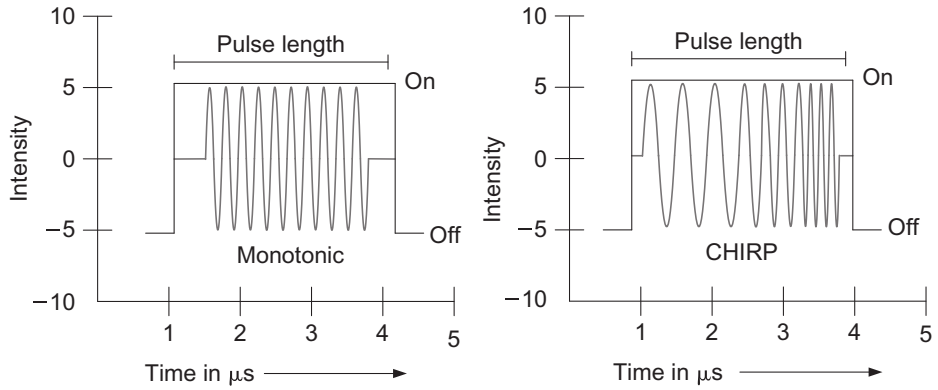


FIGURE 15.22

Comparison of monotonic (left) versus CHIRP (right) sonar techniques.

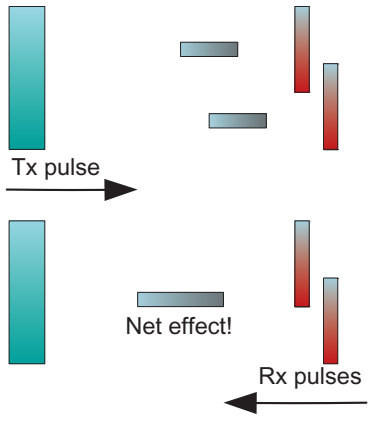


FIGURE 15.23

Example of inability of monotonic sonar to distinguish between close proximity targets.

Using the example above, if two targets are less than 37.5 mm apart, they cannot be distinguished from each other. The net effect is that the system will display a single large target, rather than multiple smaller targets.

CHIRP signal processing overcomes these limitations. Instead of using a burst of a single carrier frequency, the frequency within the burst is swept over a broad range throughout the duration of transmission pulse. This creates a "signature" acoustic pulse; the sonar knows what was transmitted and when. Using "pattern-matching" techniques, it can now look for its own unique signature being echoed back from targets.

In a CHIRP system, the critical factor determining range resolution is now the bandwidth of the CHIRP pulse. The CHIRP range resolution is given by:

$$\textit{Range resolution} = 2 \times \textit{velocity of sound bandwidth}$$

The bandwidth of a typical commercial CHIRP system is 100 kHz.
With VOS in water 1500 m/s (typical), our new range resolution = 7.5 mm—a theoretical improvement by a factor of 5!
This time, when two acoustic echoes overlap, the signature CHIRP pulses do not merge into a single return. The frequency at each point of the pulse is different and the sonar is able to resolve the two targets independently (Figure 15.24).
The response from the "pattern-matching" algorithms in the sonar results in the length of the acoustic pulse no longer affecting the amplitude of the echo on the sonar display. Therefore, longer transmissions (and operating ranges) can be achieved without a loss in range resolution.
Additionally, CHIRP offers improvements in background noise rejection, as the sonar is only looking for a swept frequency echo and removes random noise or out-of-band noise.

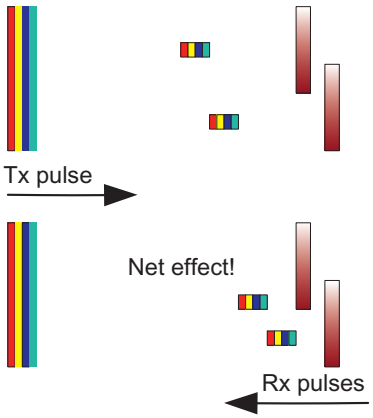


FIGURE 15.24

CHIRP sonar successfully distinguishes between close proximity targets.

In summary, CHIRP techniques provide the following advantages:

- Greatly improved range resolution compared to fixed-frequency sonars
- Larger transmission pulse lengths for increased operating ranges
- Improved discrimination between closely spaced targets
- Improved noise rejection and SNRs
- Reduced power consumption, from high-speed digital circuitry

15.2.5.2 Acoustic lens sonar

For an explanation of acoustic lens sonars, the concept will first be introduced followed by an example of a commercially available product using this technology.

From Loeser (1992) comes a general explanation of acoustic lens technology:

> *Acoustic lenses simplify the beam-forming process. The liquid lens is a spherical shell, filled with a refractive medium, that focuses sound energy in the same manner an optical lens focuses light energy. Sound waves incident on the lens are refracted to form a high-intensity focal region. The refraction is caused by a difference in acoustic wave speed in the lens media and surrounding water. The focusing ability is set by its diameter as measured in wavelengths for the frequency of interest.*
>
> *A single hydrophone located in the focal region of the lens forms a highly directive beam pattern without the necessity of auxiliary beam-forming electronics.*

The Sound Metrics name for their acoustic lens products are MIRIS and DIDSON (the latest iteration of this product series is the ARIS), referenced here as an example of the acoustic lens technology (Figure 15.25).



FIGURE 15.25

The DIDSON acoustic lens sonar.

MIRIS and DIDSON use acoustic lenses to form very narrow beams during transmission of pulses and reception of their echoes. Conventional sonars use delay lines or digital beam-forming techniques on reception and generally transmit one wide beam on transmission that covers the entire field of view. Acoustic lenses have the advantage of using no power for beam forming, resulting in a sonar that requires only 30 W to operate. A second advantage is the ease to transmit and receive from the same beam. The selective dispersal of sound and two-way beam patterns make the images cleaner due to reduced acoustic crosstalk and sharper due to higher resolution.

15.2.5.2.1 Lenses

Figure 15.26 shows a DIDSON sonar with the lens housing removed. The acoustic lenses and transducer array are shown above the electronics housing. The small cylinder under the side of the lens housing contains the focus motor and mechanism that moves the second lens forward and aft. This movement focuses the sonar on objects at ranges from 1 to 40 m. The front lens is actually a triplet made with two plastic (polymethylpentene) and one liquid (3M FC-70) component. The plastic lenses as well as the transducer array are separated by ambient water when the lens system is submerged. Optical lens design programs determined the lens curvatures. The designs were analyzed by custom software to evaluate the beam patterns over the field of view of interest.

15.2.5.2.2 Transducer array

The DIDSON and MIRIS transducer arrays are linear arrays. DIDSON has 96 elements with a pitch of 1.6 mm and a height of 46 mm. The elements are made with PZT 3:1 composite material constructed by the dice-and-fill method. The 3:1 composite provides a wide bandwidth, allowing DIDSON to operate at 1.8 or 1.0 MHz, the upper and lower ends of the transducer passband. The composite also allows the transducers to be curved in the height direction to aid in the formation of



FIGURE 15.26

DIDSON internal workings.

the elevation beam pattern. All 96 elements are used when operating at 1.8 MHz. Only the "even" 48 elements are used when operating at 1.0 MHz.

15.2.5.2.3 Beam formation

Figure 15.27 shows a ray diagram of the MIRIS system. A plane wave entering the left side through the front triplet L1 and single lenses L2 and L3 is focused to a line perpendicular to the page at the transducer T. If the normal to the plane wave is perpendicular to the front lens surface at the center, the acoustic line is formed at 0° in the diagram. If the normal is 9° off from perpendicular, the line is formed at 9° in the diagram. When a focused line of sound coincides with a long, thin transducer element, the acoustic energy is transformed into electrical energy and processed. The DIDSON beam former loses approximately 10 dB in sensitivity each way with beams 15° off-axis. Even with this reduction, DIDSON images fill the 29° field of view as shown in Figure 15.28. The average beam width in the vertical direction for both MIRIS and DIDSON is 14° (one-way). The lenses form the horizontal beam width and the curved transducer element forms the vertical beam width.

15.2.5.2.4 Image formation

The sonar transmits a short pulse and then receives its echo as it sweeps along the stripe. The echo amplitude varies in time as the reflectance varies with range along the insonified surface. Echoes from 96 adjacent lines, which together map the reflectance of the insonified sector-shaped area, are used to form a DIDSON image.

The difference between optical video and images from these sonars is more than the usage of light or sound. The sonar must be oriented to project beams with a small grazing angle to the surface of interest. The resulting image appears to be viewed from a direction perpendicular to the surface and the shadows indicate a source off to the side. Optical video could image a surface with the camera view perpendicular to the surface. If the sonar beams were perpendicular to the surface, the resulting image would show a single line perpendicular to the center beam axis. The line would be located at the range the surface is from the sonar. The sonar images are formed with "line-focused"

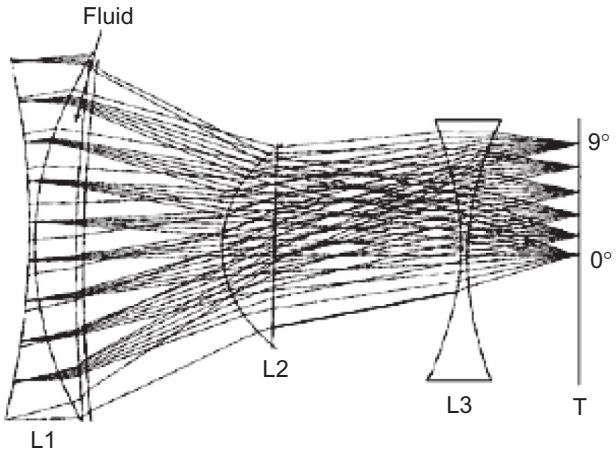


FIGURE 15.27

MIRIS internal workings.



FIGURE 15.28

DIDSON beam forming.

beams that provide good images in many conditions, but not in all conditions. If objects were at the same range in the same beam but at different elevations, this type of imaging could not sort them out. An example would be trying to image an object in a pile of debris on the ocean floor. If the object were imbedded in the pile, the acoustic images from MIRIS and DIDSON would be confusing. Video using "point-focused" optics could meaningfully image the object embedded in the pile as long as it was not totally covered.

Fortunately, the great majority of imaging tasks do not have multiple objects in the same beam, at the same range, but at different elevations. In most cases, MIRIS and DIDSON provide unambiguous, near photographic-quality images. In the newest generation of lens sonar from Sound Metrics, the ARIS sonar features a higher number of beams along with better processing power and data capture/manipulation through improvements in technology.

15.3 Sonar techniques

15.3.1 Using an imaging sonar on an ROV

The imaging sonar is a useful addition to a positioning system on an ROV. Without an imaging sonar, an ROV operator must rely on flying the submersible underwater to bring new targets into view. With an imaging sonar, instead of traveling, it is more useful to spend some time with the vehicle sitting on the bottom while the sonar scans the surrounding area. Scanning a large area takes only a short time. The vehicle pilot can quickly assess the nature of the surrounding area, thereby eliminating objects that are not of interest. The ability to "see" a long distance underwater allows the pilot to use natural (or man-made) features and targets as position references (Figures 15.29 and 15.30).

If the ROV pilot is searching for a particular object, recognition can take place directly from the sonar image. In other cases, a number of potential targets may be seen. A pilot can sharpen sonar interpretation skills by viewing these targets with the vehicle's video camera in order to correctly identify/characterize them.



FIGURE 15.29

Sonar scan of a crab trap with imaging sonar.

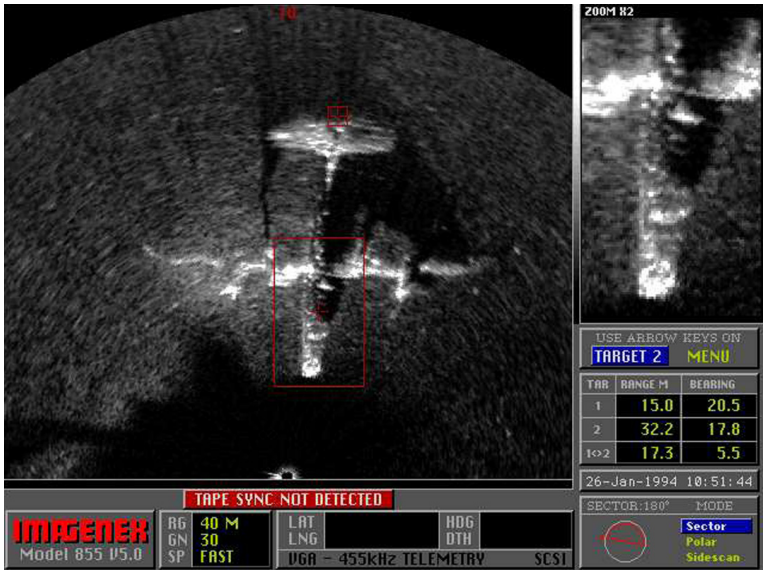


FIGURE 15.30

Convair PB4Y-2 privateer in Lake Washington, WA.

A word of caution regarding mechanically scanning sonar systems mounted on ROVs (especially OCROVs): An image is generated as the sonar transducer is rotated around its axis. If the sonar platform, that is, the submersible, is moved before the image is allowed to generate, "image smear" will occur (Figure 15.31). This phenomenon distorts the 2D display, which will not depict the correct placement of the items on the screen in *x/y* perspective.

15.3.2 Technique for locating targets with ROV-mounted scanning sonar

The accepted technique for locating the identifying items of interest insonified by scanning sonar is a four-step process:

1. Place the vehicle in a very stable position on or near the bottom to allow generation of a 360° (or wide angle) image. Depending upon the range, sector angle, and scan speed selected, the image may take up to 30 s to generate (Figure 15.32).
2. Identify the relative bearing of the item of interest.
3. Turn the vehicle to place the item at a zero bearing (in line with the bow of the vehicle), and then narrow the sector scan of the sonar system to approximately 45°. By narrowing the sector, the sonar scans only the area of the target to maintain contact.
4. Maintain contact with the item on sonar as the vehicle is driven forward toward the target until the item is in view.

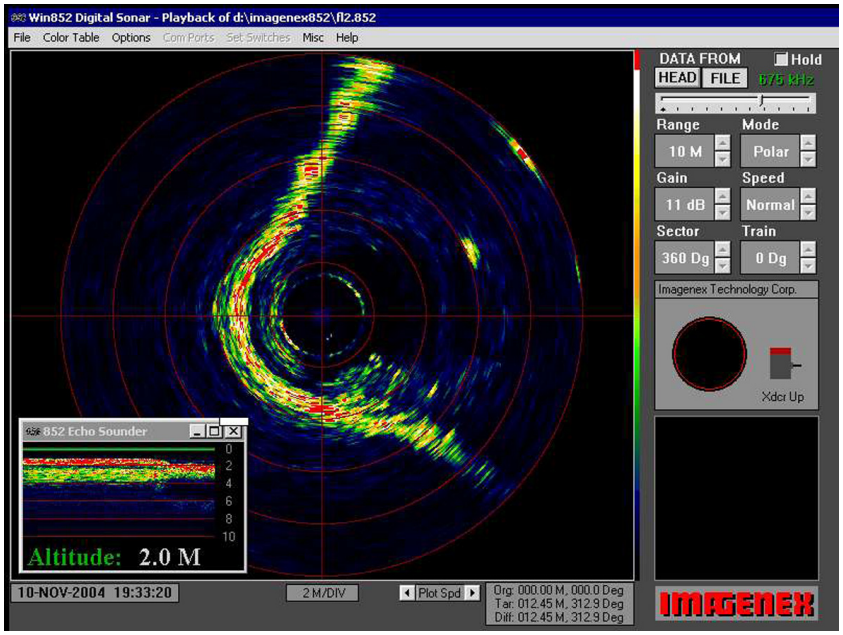


FIGURE 15.31

This image displays the sonar image of a straight wall distorted due to movement of the submersible before the image was allowed to generate.

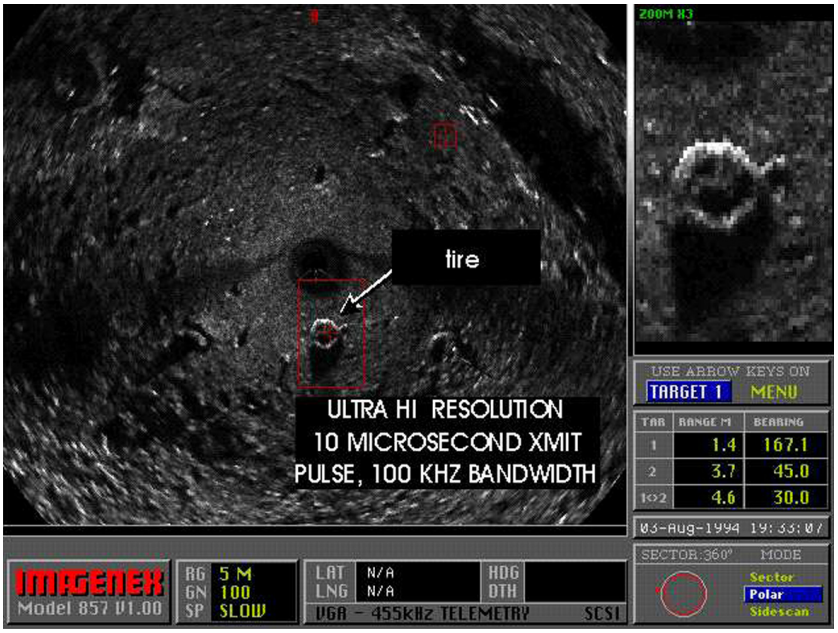


FIGURE 15.32

Ultrahigh-resolution sonar image.

15.3.3 Interpretation of sonar images

In many cases, the sonar image of a target will closely resemble an optical image of the same object. In other cases, the sonar image may be difficult to interpret and unlike the expected optical image.

The scanning process used to create a sonar image is different from the process used by the human eye or a camera to produce optical images. A sonar image will always have less resolution than an optical image, due to the nature of the ultrasonic signals used to generate it. Generally, rough objects reflect sound well in many directions and are therefore good sonar targets. Smooth angular surfaces may give a very strong reflection in one particular direction, but almost none at all in other directions (Figure 15.33). They can also act as a perfect mirror (so-called specular reflectors), reflecting the sonar pulse off in unexpected directions, never to return. This happens to people visually when they see an object reflected in a window. The human eye deals with such reflections daily, but it is unexpected to see the same thing occur with a sonar image.

As with normal vision, it is useful to scan targets from different positions to help identify them. A target unrecognizable from one direction may be easy to identify from another. It is important to note that the ranges shown to the targets on the sonar image are "slant" ranges. Usually the relative elevations of the targets are not known, only the range from the transducer. This means that two targets, which are displayed in the same location on the screen, may be at different elevations. For example, you might see a target on the bottom and a target floating on the surface in the same place (Figure 15.34).

By analyzing the shadows, an estimation of the height of objects above the bottom can be ascertained. An example of this calculation is shown in Figure 15.35.

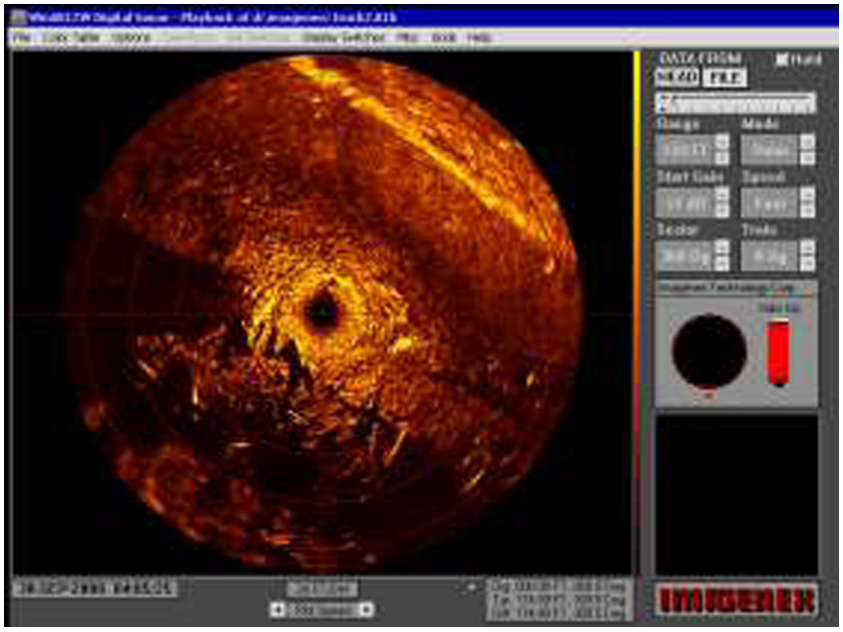


FIGURE 15.33

On the left side of the sonar display is a Ford F-150 pickup truck in 40 ft of water completely reflecting sonar signal (only shadow is recognized).

The diagrams in this chapter are examples of the sonar scanning process. Studying the diagrams will help users to better understand the images seen. A basic knowledge of this process will help users interpret what otherwise might be confusing images.

15.4 New and emerging technologies

Since the publication of the first edition of this text, a new technology—3D imaging sonar—has provided some revolutionary capabilities. This technology takes place in both hardware as well as software, allowing for full 3D rendering of *in situ* targets for various applications including metrology and modeling.

15.4.1 Image capture

The five methods for capturing 3D acoustic data are as follows:

1. Holding the platform stationary and then swinging the 2D beam projector to capture the third dimension (Figure 15.36)
2. Fixing the beam stationary relative to the platform and then moving the platform to capture the third dimension (Figure 15.37)
3. Rendering with a wider 3D beam (Figure 15.38) for full volumetric image capture
4. Separate two (or more) individual fixed projectors and render simultaneously (Figure 15.39) as the projectors are simultaneously moved (i.e., a small variation on item 2 above)
5. Physically rolling (rotating) the 2D beam/sonar housing, thus rendering the third dimension by capturing the vertical orientation (Figure 15.40)

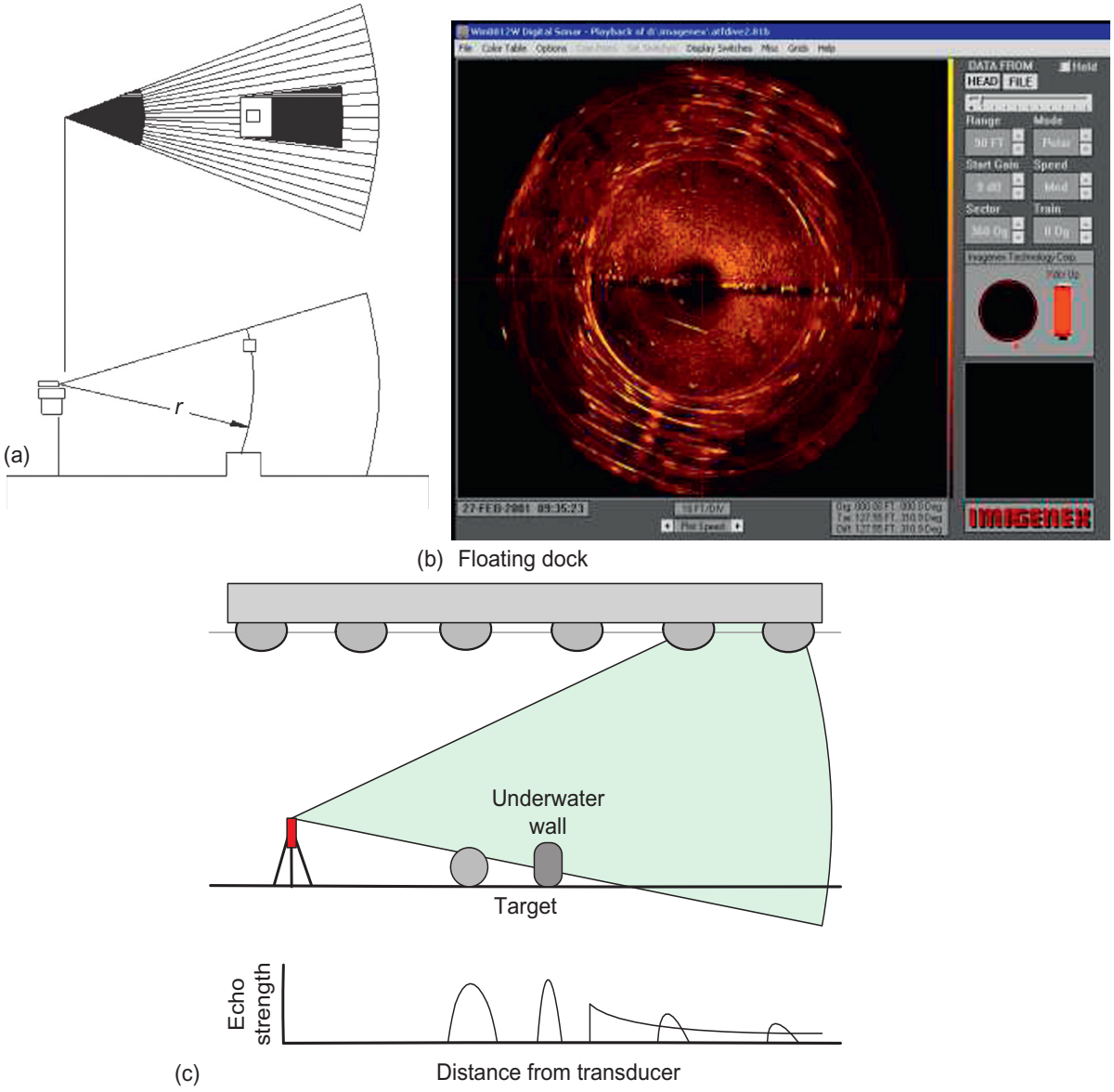


FIGURE 15.34

(a) Graphic illustration of two targets (one on surface and one on bottom) displaying colocation, (b) as depicted on the sonar display and (c) as imaged compared to signal strength.

15.4.2 Image rendering

Rendering of both 2D and 3D images from acoustic capture is done one of two ways:

1. With 2D data rendering, a bottom scene of a small physical area is scanned and imaged from several locations within the scene. Then each scan is laid on a digital raster illustration package

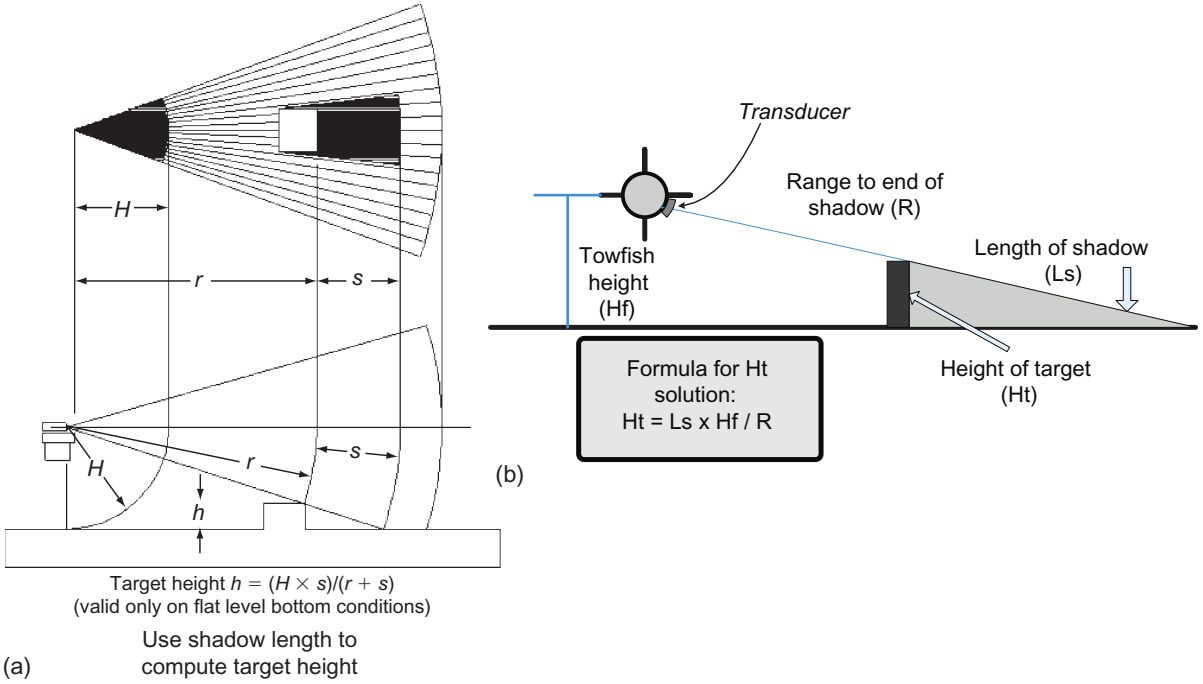


FIGURE 15.35

Target height calculations (a) with transducer graphically depicted and (b) shown with tow fish superimposed.

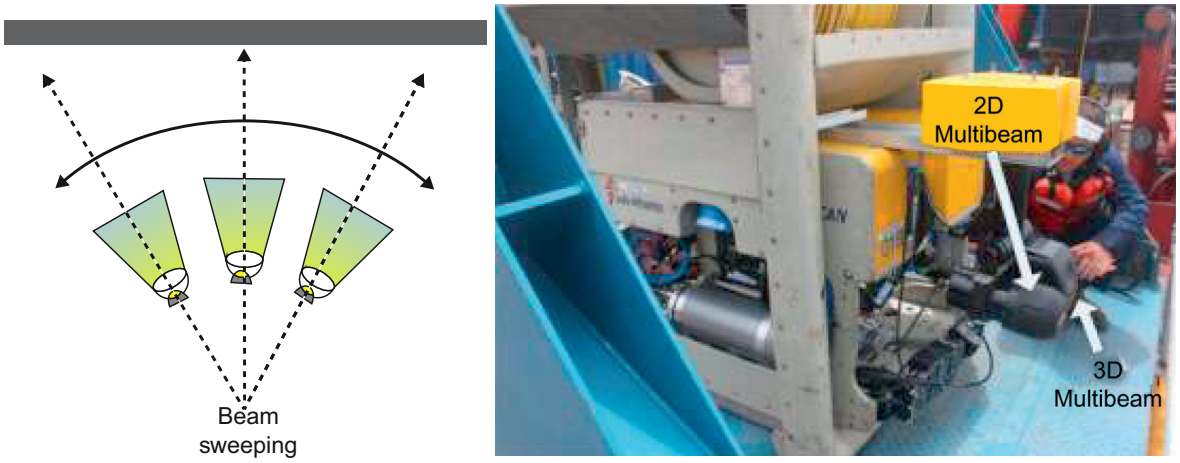


FIGURE 15.36

Concept of electronically sweeping multibeam and commercial application.

(*Courtesy SeaTrepid.*)

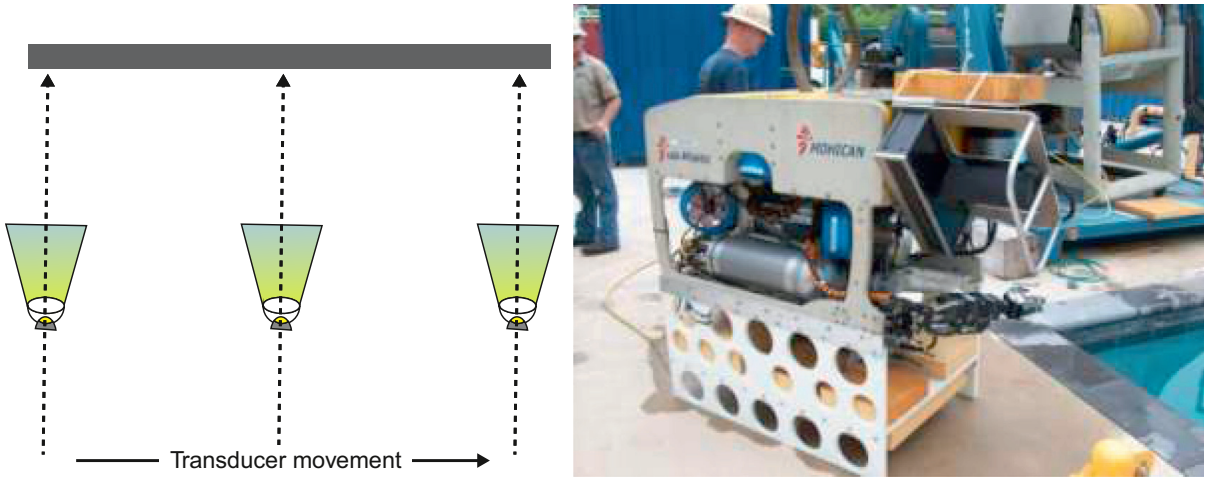


FIGURE 15.37

Concept of rendering 3D data by moving the sonar projector along with a fixed multibeam attached to an MSROV.

(*Courtesy SeaTrepid.*)

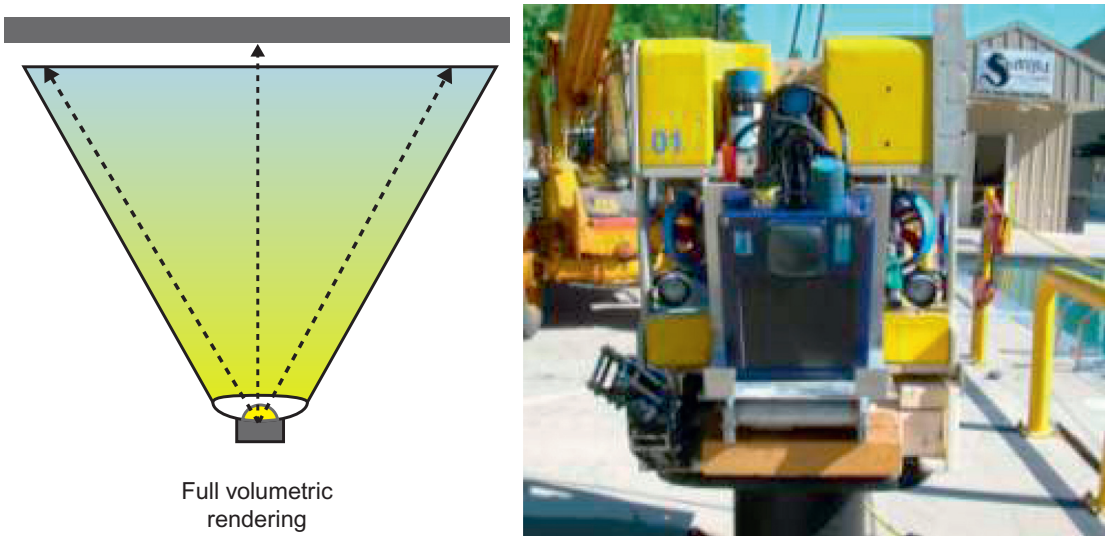


FIGURE 15.38

Concept of rendering 3D data through full volumetric coverage by sonar projector along with a Coda Octopus Echoscope integrated into an MSROV.

(*Courtesy SeaTrepid.*)

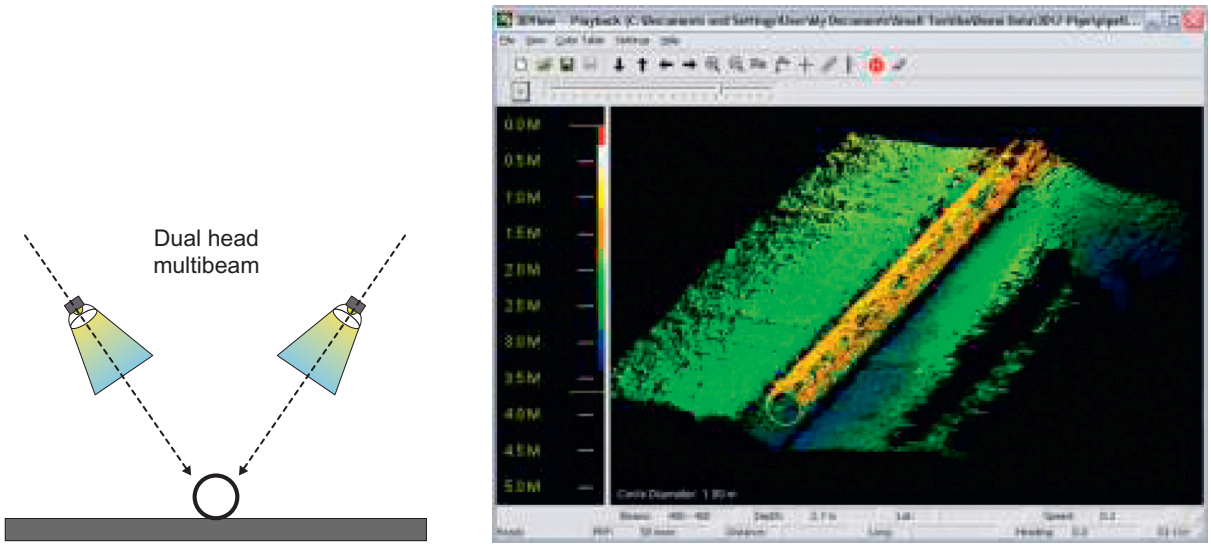


FIGURE 15.39

Concept of dual head multibeam and sample pipeline rendering.

(*Courtesy Imagenex.*)

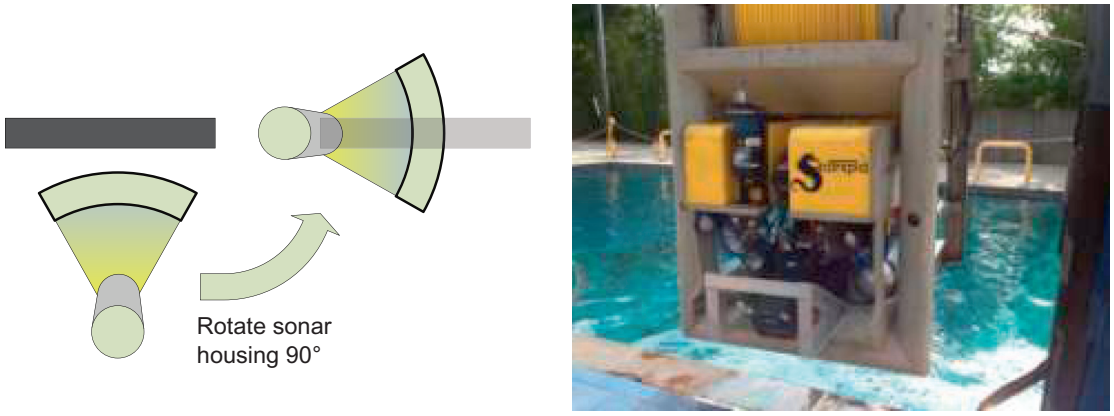


FIGURE 15.40

Concept of rotating a 2D sonar from horizontal to vertical orientation along with a Sound Metrics ARIS with X2 rotator integrated into an MSROV.

(*Courtesy SeaTrepid.*)

(e.g., Adobe Photoshop or similar). The individual items within the scene that are common between scans are then linked/stitched together to form a mosaic.

2. With 3D data rendering, the same series of scene capture is placed in a mosaic format. The difference between the 2D and 3D mosaic is that the 3D data is plotted into a 3D rendering software package (e.g., Leica Cyclone or similar) for the *x*/*y*/*z* point coordinates, thus producing a 3D "point cloud" of individual echoes (see Figure 23.8 in Chapter 23). This is particularly useful for modeling of *in situ* structures based upon real data for metrology purposes.